

Contents

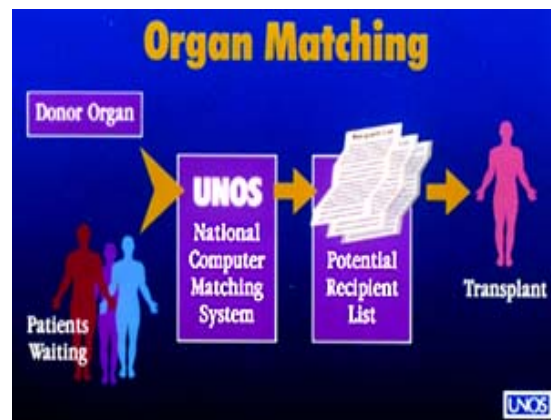
1. Introduction-----	1
2. Assumptions-----	3
3. Symbols and Definitions -----	3
4. Solutions-----	4
4.1 Task 1-----	4
4.1.1 Problem Analysis-----	4
4.1.2 Model design-----	7
4.1.3 Potential Bottlenecks Analysis-----	10
4.1.4 Further Discussion-----	10
4.1.5 Example analysis-----	11
4.1.6 Model improvement-----	12
4.2 Task 2-----	13
4.2.1 Evaluating the OPTN policy-----	13
4.2.2 One-page Report-----	17
4.3 Task 3-----	18
4.3.1 Problem Analysis-----	18
4.3.2 Model Design-----	19
4.3.3 Simulations example-----	23
4.3.4 Conclusion-----	25
4.4 Task 4-----	26
4.4.1 Problem Analysis-----	26
4.4.2 Model Design-----	26
4.4.3 Solution and result-----	30
4.5 Task 5-----	32
4.5.1 Analysis on Ethical and Political issues-----	32
4.5.2 Changing and Ranking order the criteria-----	35
4.5.3 Modifying the Models in Task 3 and Task 4-----	39
4.5.4 Organ can be sold or not?-----	40
4.5.5 One-page Report-----	43
4.6 Task 6-----	44
4.6.1 Risks Analysis-----	44
4.6.2 Model design-----	45
4.6.3 Modify Models-----	47
4.6.4 Appeal to more donors-----	48
5. Strengths and Weaknesses-----	49
6. References-----	50
7. Appendix -----	51
7.1 Appendix I -----	51
7.2 Appendix II -----	57

Organ Transplant: The Kidney Exchange Problem

Introduction

Transplant Network:

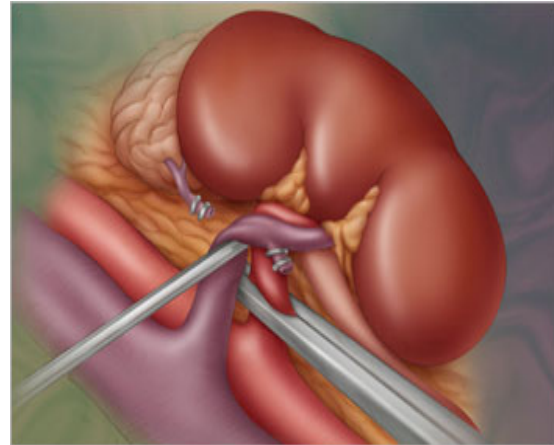
Despite the continuing and dramatic advances in medicine and health technology, the demand for organs for transplantation drastically exceeds the number of donors. To help this situation, US Congress passed the National Organ Transplant Act in 1984, establishing the Organ Procurement and Transplantation Network (OPTN) to match organ donors to patients with organ needs. Even with all this organizational technology and service in place, there are nearly 94,000 transplant candidates in the US waiting for an organ transplant and this number is predicted to exceed 100,000 very soon. The average waiting time exceeds three years—doubles that in some areas, such as large cities. Organs for transplant are obtained either from a cadaver queue or from living donors. The keys for the effective use of the cadaver queue are cooperation and good communication throughout the network. The good news is that the system is functioning and more and more donors (alive and deceased) are identified and used each year with record numbers of transplants taking place every month. The bad news is that the candidate list grows longer and longer. Some people think that the current system with both regional and national aspects is headed for collapse with consequential failures for some of the neediest patients. Moreover, fundamental questions remain: Can this network be improved and how do we improve the effectiveness of a complex network like OPTN? Different countries have different processes and policies, which of these work best? What is the future status of the current system?



Focusing on Kidney Exchange:

Kidneys filter blood, remove waste, make hormones, and produce urine. Kidney failure can be caused by many different diseases and conditions. A kidney transplant can come from the cadavers of an individual who agreed to donate organs after death or from a live donor.

Two genetic characteristics play key roles in the possibility and success of a kidney transplant. The first of these is the ABO blood-type: In the absence of other complications, kidneys can be successfully transplanted if they do not contain blood proteins that are foreign to the patient. So type O kidneys can be transplanted into any patient; type A kidneys can be transplanted into type A or type AB patients; type B kidneys can be transplanted into type B or type AB patients; and type AB kidneys can only be transplanted into type AB



patients. The second genetic characteristic that affects the success of a kidney transplant is the tissue type, also known as human leukocyte antigens (HLA) type: HLA type is a combination of six proteins, two of type A, two of type B, and two of type DR. the transplantation is not carried out if there is a mismatch. Several factors such as patient race, patient age, patient health condition (more specifically previous history of diabetes), patient gender, donor type (cadaver or live), donor age, HLA mismatches can be involved from the medical perspective.

Recently, however, in a relatively small number of cases, additional possibilities have been utilized when a transplant from a live donor and the intended recipient is infeasible. One of these, called a paired exchange, involves two patient-donor couples, for each of whom a transplant from donor to intended recipient is infeasible, but such that the patient in each couple could feasibly receive a transplant from the donor in the other couple. This pair of couples can then exchange donated kidneys. Compared to receiving cadaver kidneys at an unknown future time, this improves the welfare of the transplant recipients (and accomplishes the aim of each patient's donor). In addition, it relieves the demand on the supply of cadaver kidneys, and thus potentially improves the welfare of those patients on the cadaver queue. A small number of these two-couple operations have been done, and the transplantation community has issued a consensus statement declaring them to be ethically acceptable.

Another possibility is an indirect exchange involving an exchange between one immunological infeasible patient-donor couple, and the cadaver queue. In this kind of exchange, the patient in the couple becomes the highest priority person on the cadaver queue, in return for the donation of his donor's kidney to someone on the queue.³ This improves the welfare of the patient in the couple, compared to having a long wait for a suitable cadaver kidney, and it clearly benefits the recipient of the live kidney, and others on the queue who benefit from the increase in kidney supply due to an additional living donor. .

In contrast to the system for cadaver organs, and despite the growing interest in at least small scale exchanges involving living donors, there is no national system, or even an organized system at any level, for managing exchanges of kidneys from live donors. However, individual hospitals are beginning to think about larger scale living donor exchanges.

Kidney Exchange also involves many other aspects. An idea that originated in Korea is that of a kidney exchange system, which can take place either with a living donor or with the cadaver queue. We have the following main considerations about the medical matching, the kidney exchange problem, criteria on the ethics and policy issues, etc.

Assumptions

- The statistic data is true and reliable;
- In task 3, the influence of ethical and political issues is neglected;
- In task 4, we suppose while a failed transplant results in immediate death;
- Considering the deceased donors can donate two kidneys, so we suppose the donation bases on the kidney unit.

Symbols and Definitions

OB	Organ Bank
UKT	U.K Transplant
R-C	Rational–Cautious
pmp	per million population
ODR	Organ Donor Register
ESRD	end-stage renal disease
NHS	National Health Service
NBA	National Blood Authority
HLA	Human Leucocyte Antigen
CMS	Centers for Medicare Services
BTS	British Transplantation Society
TTCC	Top Trading Cycles and Chains
NKRF	National Kidney Research Fund
OPO	Organ Procurement Organization
UNOS	United Network of Organ Sharing
OPTN	Organ Procurement and Transplantation Network

Solution

Task 1

Problem Analysis

In the task 1 of our study, we are required to build a mathematical model for the generic US transplant network according to the policy description of the Organ Procurement and Transplantation Network (OPTN) given on the OPTN website. We are also supposed to do some analysis of a series of related issues following the model. In our model designing, we can firstly do some research and analysis of the OPTN the relevant policy, and then search the organ donation data (primarily renal transplantation) on the Internet, thus a network model including the input, output and the organ transport database can be formed. The core of the network is the queue matching problem that is how an Organ Procurement Organization (OPO) deals with the input and output data information and properly distributes the available organs to the candidates on the waiting list. After the establishment of the model, we can do some analysis and testing about the problem and give insight into the following: Where are the potential bottlenecks for efficient organ matching? If more resources were available for improving the efficiency of the donor-matching process, where and how could they be used? Would this network function better if it was divided into smaller networks (for instance at the state level)? In order to answer this question, we can consider these issues through theoretical analysis, and then use Monte Carlo simulation method to reach a simple example of proof. Finally, the model can be appropriately improved by a changed idealistic strategy. So that we make the system more effective by saving and prolonging more lives.

OPTN policy analysis

The OPTN policies and bylaws govern the procedural aspects of policy development, allocation of donated organs, and the collection of transplant data nationwide. The public is included in the process through solicitation of feedback via the public comment process. From the OPTN's policy descriptions ^[1], we can obtain that the Organ Procurement and Transplantation Network mainly contains an OPO, transplant centers, transplant Program, Waiting List, Match System, Alternative allocation/distribution System, as well as the Sharing Arrangement and Sharing Agreement.

Consider this very comprehensive policy, where many of the details of a more detailed explanation. The OPO is an organization, accepted as a Member, and authorized by the Centers for Medicare Services (CMS) to procure organs for transplantation. The policy descriptions also give out the blood type matching of the organ transplant, the description of the degree of transplantation matching. In addition, some criteria of many aspects are presented.

Given the key issue of the organ transplants problem is waiting list matching algorithm, we put the emphasis on the waiting list analysis and matching system part. The Waiting List is the computerized list of candidates who are waiting to be matched with specific donor organs in hopes of receiving transplants. Waiting List candidates are registered on the Waiting List

by member transplant centers. The candidate's transplant program shall be responsible for ensuring the accuracy of candidate ABO data on the waiting list. Each transplant program shall implement and operate procedure for providing on-line verification of a candidate's ABO data on the waiting list against the source document by an individual other than the person initially entering the candidate's ABO data in UNetSM. The transplant program shall maintain records documenting that such separate verification of the source document against the entered ABO has taken place and make such documentation available for audit. Upon entry of the candidate's waitlist data, the candidate will be added to the waitlist but will not be listed as an active candidate until separate verification of the candidate's ABO data has taken place. The Match System is the computerized algorithm used to prioritize candidates waiting for organs. It eliminates potential recipients whose size or ABO type is incompatible with that of a donor and then ranks those remaining potential recipients according to the ranking system approved by the Board.

Based on the above analysis, as well as some of the other details to deal with, we can establish a corresponding mathematical model behind to simulate the reality of organ transplantation matching mechanism. Once our model take use of some part of this policy, we will make corresponding labels.

Data collection and analysis

We can obtain the data which is involved with the status of the American Organ transplant from the data banks.^{[2][3]} We have collected the demand of the various organs in United States to date, the annual donors, transplants and the demand (Here taking the kidney for example, by years 1995-2006).

Table 1: The demand of the various organs in United States to date

Based on OPTN data as of February 2, 2007	All Organs	Kidney	Liver	Pancreas	Kidney / Pancreas	Heart	Lung	Heart / Lung	Intestine
Total	94,545	69,983	16,989	1,745	2,389	2,858	2,860	135	239
Percent	1	74.021%	17.969%	1.846%	2.527%	3.023%	3.025%	0.143%	0.253%

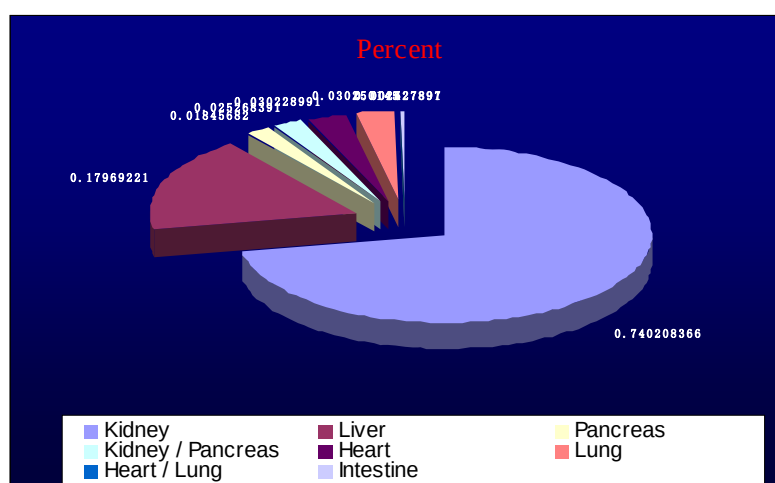


Figure 1: The percent of the various organs

From the above table1 and figure1, we can see that the kidney accounts for 73% in the total of the organ transplants. It accounts for a very large proportion as a most important organ which can be transplanted. Therefore, we only need to discuss the status of the kidney transplant here, being able to achieve the analysis and research on the organ transplant.

Table 2: The annual donors, transplants and the demand of the kidney

	Donor			Transplant			Waiting List		
Year	Deceased Donors	Living Donors	All Donor Types	Deceased Donors	Living Donors	All Donor Types	Waiting List Additions	Waiting List removals	All Waiting List
1995	5,001	3,392	8,393	7,694	3,387	11,081	17,270	28,067	45,337
1996	5,036	3,678	8,714	7,730	3,668	11,398	17,735	28,873	46,608
1997	5,083	3,933	9,016	7,774	3,927	11,701	18,437	30,272	48,709
1998	5,339	4,421	9,760	8,032	4,419	12,451	19,497	32,460	51,957
1999	5,386	4,724	10,110	8,043	4,717	12,760	21,868	35,658	57,526
2000	5,489	5,493	10,982	8,124	5,488	13,612	22,373	38,179	60,552
2001	5,528	6,038	11,566	8,230	6,035	14,265	22,517	39,293	61,810
2002	5,638	6,240	11,878	8,539	6,240	14,779	23,641	41,811	65,452
2003	5,753	6,473	12,226	8,667	6,470	15,137	24,694	42,686	67,380
2004	6,325	6,647	12,972	9,357	6,647	16,004	27,290	45,615	72,905
2005	6,700	6,570	13,270	9,913	6,568	16,481	29,166	50,114	79,280
2006	6,609	5,913	12,522	9,807	5,914	15,721	29,824	49,760	79,584
To Date	97,820	80,121	177,941	152,678	80,077	232,755	239,793	442,264	682,057

According to the above data, we can get the figures as follow:

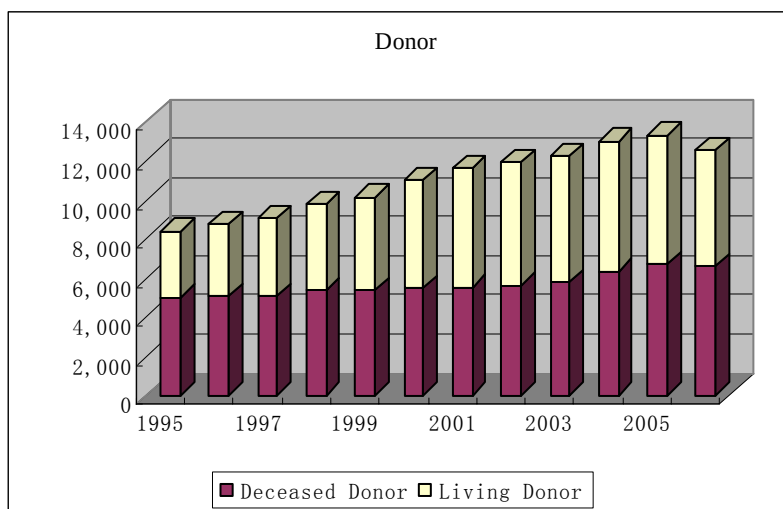


Figure 2: The quantity of Deceased and Living Donors

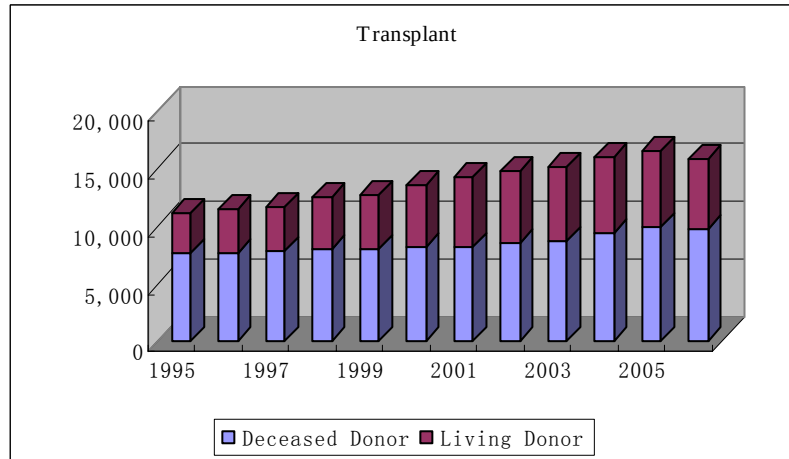


Figure 3: The quantity of Deceased and Living Transplant

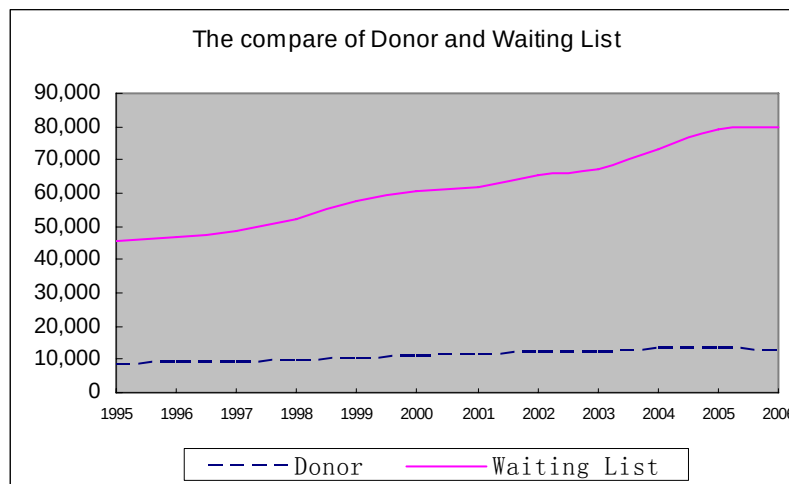


Figure 4: The compare of Donor and Waiting List

From the above table2 and figure2-4, we can conclude some points as follow:

- In the organ donations, the number of organs from deceased donors is larger than which comes from the Living Donors;
- The rapid growth takes place in the number of organ donations from the Living Donors;
- Donors are less than the demand, and the growth rate of the demand is much higher than the donors.

Model design——OB Network Model

As we have analyzed in the policy part, the organ transplant problem can be described as a waiting list Matching problem. By involving the statistical data information about the candidates and the donors, Status of the U S's organ Transplant can be clearly obtained, and then we can built a mathematical model – a network based on the matching of organ transplant. This network consists of three parts, namely, the Input, Output and OB (organ bank) part. See the network as follow ^[4]:

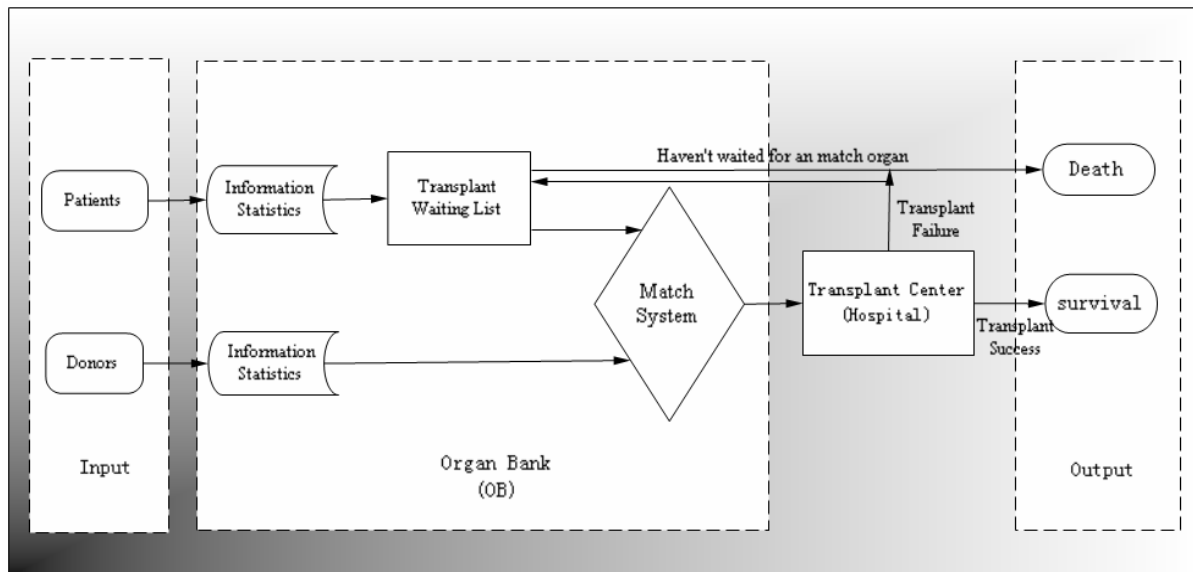


Figure 5: Network Model flowchart

- **Input part**
This part includes the data information of the candidates who need an organ transplant and the organ donors. The donors can be divided into living donors and the cadavers or be classified into immediate donors and potential donors. The data information has to include the basic items such as the blood type, HLA, waiting time and so on. If we consider the issue in detail, some other items like the candidate's age, degree of medical urgency should be taken into account to modify our model.
- **OB(organ bank)**
OB(organ bank), an information-processing site, Its main functions include information statistics on the patients and donors, information retrieval, transplant waiting queue and matching system parts. The core of the OB system is that the information will be placed into a waiting sequence and an available list, and a suitable matching algorithm should be found for the system.
- **Output part**
Show the success of organ transplant, the matching results and matching quality should be available in this part.

The network processes as follow:

- 1) The patient who needs for the organ transplant apply for Organ Bank and input his relevant information, including the application time, medical condition, the size of his organ which need to be transplanted, Age, Sex, HLA, Blood Type and so on;
- 2) The patient (information) is put into the Transplant Waiting List. Then determining its position in the queue through synthesizing of all the information. Along with the cumulative waiting time, he will also become the location of the front line;
- 3) The donor apply for Organ Bank and input his relevant information, including the donated time, the size of his organ, Age, Sex, HLA, Blood Type and so on;
- 4) When a donor arises, the patient who located in front of the queue and the donor's information will be put into the Match System. If the two can match, then enter the Transplant Center (Hospital) for transplantation; If the patient has not found his matching organ in a very long period of time, he will be removed from the queue or directly death;

- 5) If transplantation succeed, the patient survival; Transplant failure, the patient returns to the queue to wait a next matching organ or directly death.

A very specific image about our model is determined by the above network flowchart. The input-output relations of the three parts are clearly described. Particularly in the match system, the blood type matching, HLA, some relevant test standards can be taken into account.

Definition and notation

We let $\omega_1 \ \omega_2 \ \omega_3$ be the given weight, they can be modified for different objects, and the w_{ij} stands for the matching relation between the candidate p_i and the donor p_j , it is a binary variable. p_{ij} is the degree of a organ transplant matching, it can be calculated by the Point Assignment Systems Point Assignment Systems in the policy description part of the policy. c_{ij} can be constructed to describe the energy consumption of the transplant program, it consist of telecommunications outages and the transport consumption.

Objective Function

In the OB system, according to the queue waiting situation, by introducing relevant binding targets and Constraints, the corresponding matching rules can be determined to optimize the matching quality and number as follows:

$$\begin{aligned} \max mize \quad & \omega_1 \cdot \sum_{i,j \in list} w_{ij} + \omega_2 \cdot \sum_{i,j \in list} p_{ij} \cdot w_{ij} \\ \min mize \quad & \omega_3 \cdot \sum_{i,j \in list} c_{ij} \cdot w_{ij} \end{aligned}$$

Constraints

Member Rights and Obligations, membership in the Corporation is voluntary; rights and obligations of the candidate p_i and the donor p_j are referenced by the OPTN policy.

- 1) For each candidate, its waiting time should meet a certain time windows:

$$T_{i\min} \leq T_i \leq T_{i\max};$$

- 2) If $w_{ij}=1$, p_{ij} must be great than its lower limit, in order to guarantee the quality of the match.
- 3) For the candidate p_i and the donor p_j , the matching rules should hold, including the blood type, the HLA and some other items.

Different policies may determine the constraints, and will also lead to a different matching efficiency. In actual operation, the weight $\omega_1 \ \omega_2 \ \omega_3$ and the description of the objects can be modified in a reality way, Therefore, we can modify the model for certain amendments and improvements, which will be used in task 2 of our paper.

Organ Allocation strategy

According to the OPTN policy, the Point System allocation strategy is currently utilized by the United Network of Organ Sharing (UNOS). Besides this strategy, there are three other policies: a first-come first-transplanted system; an efficiency-based algorithm that incorporates correlates of patient and graft survival; and a distributive efficiency algorithm. Those are just two of the issues: the justice—which looks at how long a person has been on the waiting list; the utility—which seeks to make the best use of scarce resources. Compared with the Point System allocation strategy, in the efficiency and distributive efficiency algorithms, the highest priority candidate is the one with the greatest increase in quality-adjusted life expectancy if transplanted. For each candidate and organ, this was calculated as the quality-adjusted life expectancy assuming a transplant minus the quality-adjusted life expectancy assuming no transplant and no future transplant. (Because these estimates do not account for future transplants, algorithms employing more elaborate calculations might improve estimated quality-adjusted life expectancy.)

Potential Bottlenecks Analysis

Despite the matching networks can better organize the organ transplant matching program, and also ease the difficult situation of the United States for organ transplant to some extent. However, due to the complexity of the network and its own attributes of the organ transplant system, there certainly exist certain bottlenecks for the efficiency of matching organ transplant network.

- Limited resources available for organ transplant. From figure 4 above, it is true that despite more and more donors (alive and deceased) are identified and used each year, the candidate list grows longer and longer, more fast than the donor case. Limited resources will bring a fundamental limit to this problem.
- Policy constraints and matching rules in our model are too strong, the feasible solution and efficiency of the matching results are largely restrained.
- The organ procurement and transplantation problem is big second graph matching problem. As the n is very large, the network became dramatically complex and the matching optimization results can not be very good.
- Various special circumstances in the case of emergency make the network system more comprehensive and acceptable. However, the system also becomes very difficult to deal with. Thus the modification of the objects and the weight $\omega_1 \ \omega_2 \ \omega_3$ become difficult.

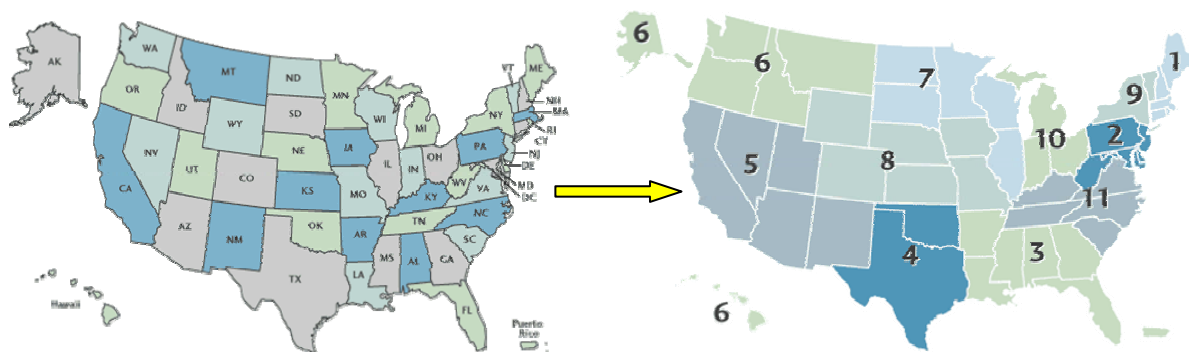
Further Discussion

Problem:

- If more resources were available for improving the efficiency of the donor-matching process, where and how could they be used?
- Would this network function better if it was divided into smaller networks (for instance at the state level)?

Analysis:

- For problem 1, when there are more resources for improving the efficiency of the donor-matching process, we could utilize these resources fully. In fact, the organ donations are far from adequate to meet the demand of patients. So we can use the Transplant Waiting List and the Match System of the Organ Bank in the network which is founded in our model. In OB, it determines each patient's position in the queue through synthesizing their information. Their information includes the application time, medical condition, the size of his organ which need to be transplanted, Age, Sex, HLA, Blood Type and so on. When there are new donors, these new resources will be used at the patients who standing in the front of the queue, so let the more patients be able to transplant the organ and save more lives.
- For problem 2, if the whole network is divided into smaller networks, we believe that will have both negative and positive aspects. The negative aspect is, when the network is divided into several smaller networks, the links between the each network will be reduced. Thus there will be possible to the case like this: a patient can not find a matching organ at his located network, but this organ exists in another network. As there is a delayed information transmission between the two networks, this patient died because of failure to transplant in good time. The positive aspect is, when the network is divided, OB will become more number. So the costs of the transport and preservation of the organ will be reduced greatly. And it also can reduce the total waiting time of the patients. For example, there are 50 states in U.S. According to the various factors just as the quantity of the donations, the number of patients on the waiting list for transplants, the geographical location, population density and so on, the U.S(the whole network) is divide into 11 regions(smaller networks), as follow:



- Unifies the above two aspects, under the circumstances of the network being divided, if there was a new organ resources, then we can achieve the maximum use of those resources, and allow more patients to obtain an organ transplant.

Example analysis ——Monte Carlo Simulation

The Monte Carlo method solves a problem by simulating directly the physical process, and is not necessary to write down the differential equations that describe the behavior of the system. This is very general and is valid not only for our real options problem as in other areas of knowledge like physics, chemistry, etc. ^{[5][6]}

Here we developed a Monte Carlo model to simulate the operations of the transplant network. It is based on the random Signal processing. In the model we define the number of

donors will be between 5000 and 10000, the number of the patients will be between 20000 and 40000. The data information of the donors and waiting candidates are randomly generated. For each candidate, we simulated the following characteristics: HLA typing, region, blood transfusions, priority. Donor characteristics include HLA typing, region, blood transfusions. The generated characteristics of all the candidates and donors are saved on two lists (the waiting list and the organ available list). The process of generating random data is referenced by the related data of the kidney transplant in OPTN.

We know that the United States was divided into 11 sub-regional networks. Then we simulate the organ matching process in the small network. New transplant candidates and donor organs were generated according to a random process with arrival rates of 60% and 15%, respectively. And then, we introduce more available resource to consider this relative situation. When running a Monte Carlo Simulation (See the specific procedures in Appendix I), we are supposed to run it enough times to have a statistically significant output. So after many times simulation under the conditions discussed above, we obtain statistical results as follows:

Table 3: The result of the Monte Carlo simulation

Not Divide	Donor	Patient	Transplant	Matching rate	Cost
1	6917	23285	3577	51.71%	13112
2	7390	25282	3841	51.98%	14458
3	8949	30540	4607	51.48%	17542
average	7752	26369	4008	51.72%	15037
Divide	Donor	Patient	Transplant	Matching rate	Cost
1	6917	23457	3690	53.35%	8388
2	7390	24703	4066	55.02%	9189
3	8949	31156	4819	53.85%	11272
average	7752	26439	4191	54.07%	9617

By analyzing the above result, we can find: When there are more donors (more resources), the number of transplant will increase obviously, and the matching rate changes only a little; When the network is divided into 11 regions (small networks), the costs of the transport and preservation of the organ will be reduced greatly.

Model improvement

In order to perfect our network system, that is to consider the issues more humane to make the system more effective by saving and prolonging more lives. We can take the following factors into consideration, in order to promote the network system and our model.

- Minimize the maximum waiting time;
- Modify the data information of the waiting list in a dynamic way;
- Build communications mechanisms with other Networks, thus achieving the sharing and mutual complement of available resources;
- Higher priority for the emergency Situations.

For every improving point, we are able to modify some of the constraints and objectives of restructuring, and then the corresponding correction model is obtained.

Task 2

Evaluating the OPTN policy

Despite the continuing and dramatic advances in medicine and health technology, the demand for organs for transplantation drastically exceeds the number of donors. To help this situation, US Congress passed the National Organ Transplant Act in 1984, establishing the Organ OPTN to match organ donors to patients with organ needs. Consider this very comprehensive policy, where many of the details of a more detailed explanation. Along with the establishment and perfection of the OPTN, the U.S has become one of the countries that do the best in the organ transplant all over the world. The U.S has already adopted a number of laws and regulations which related to organ transplants, the quantity of annual donors and transplants is the highest in the world.

However, although no significant disparities for organ transplant rates for the sickest patients exist on a regional, gender or racial basis under the current OPTN policy, the needs of the most critically ill patients are not met. In the U.S, about 68,000 patients are waiting for a kidney from a deceased donor, while each year only 10,000 are transplanted from cadavers and 6,000 from living individuals. Hence the median wait for a matching kidney is three years—unfortunately, some needy patients do not survive long enough to receive a kidney.

Moreover, there are a few disadvantages and limitations in the OPTN policy. For example, the point system allocation strategy is currently utilized by the UNOS.^[7] Those are just two of the issues: the justice—which looks at how long a person has been on the waiting list; the utility—which seeks to make the best use of scarce resources. Although the UNOS system strives to be fair, officials acknowledge disparities based on blood type and geography. This allocation strategy indicates that the more time a patient stays in the waiting queue, the more priority he will obtain. However, this will not enable to give a transplant to the patients with critical illness in time, so he has the possibility to face the death extremely. American citizens have a certain degree of the complaints and suspicious to this waiting system. In addition, when there are an enormous increasing number of patients, the organ donors are insufficient obviously. The OPTN has not very effective policies to solve this bottleneck.

Analyzing the organ transplant policies in other countries

There is a figure of Legislation, practice and donor rates in several countries, e.g., Spain, US, France, Germany, UK and so on.^[8]

Table 4: Legislation, practice and donor rates

Country	Legislation	Actual practice	Donors(pmp) annual rate
Spain	Presumed consent	Informed consent	33.8
Belgium	Presumed consent	Presumed consent and family informed	24.4
Austria	Presumed consent	Presumed consent	23.3
US	Informed consent with	Informed consent with	22.1

	required request	required request	
Ireland	Informed consent	Informed consent	21.1
Norway	Presumed consent	Informed consent	19.2
France	Presumed consent	Informed consent	18.3
Netherland	Informed consent	Informed consent	14.9
Germany	Informed consent	Informed consent	13.8
Sweden	Presumed consent	Varies, recent changes	12.8
U.K	Informed consent	Informed consent	12
Greece	Presumed consent	Informed consent	6.4

(pmp: per million population)

According to the above data, we can see that many of the European countries have the high rates of the donor, particularly in Spain. This phenomenon shows that the organ transplant is also hot in Europe. Although the relevant policies and statutes in these countries are less comprehensive than that in U.S, there still a lot what U.S could learn from. Here, we mainly analyze the organ transplant policies in Spain, U.K and Korea this three countries.

Spain

Spain has the highest at nearly 34 per million populations (pmp). Higher donation rates are often attributed to having opt-out legislation in place. However other factors such as how efficiently potential donors are realised and the number of intensive care beds are also important. We identified the following factors as being associated with high donation rates:

- An opt-out policy
- A large number of transplant centres pmp
- A high percentage of the population enrolled in university education
- And a high percentage of Roman Catholics (the Roman Catholic church had been previously cited for its support of donation).

In Spain the high donation rate is thought to be mainly due to the way organ transplantation is co-ordinated. While the legislation (presumed consent) has remained unmodified since 1979, organ donation rates increased during the 1990s. This followed the formation in 1989 of a National Transplant Organisation and a national network of specially trained and dedicated hospital physicians in charge of the whole process of organ donation. The new 'Hospital- Centre' approach now covers nearly all major Spanish hospitals. A proactive donor detection programme, systematic death audits in hospitals, economic reimbursement for hospitals and the social climate all contribute to the high donation rate. Spain also has a high number of road traffic accident deaths which raises potential donor numbers.

U.K

U.K has a comprehensive health organization; the organ transplant in U.K is also very comprehensive. Here, we only discuss the differences between the U.K policy and the OPTN policy. Then analyze the superiority of these different policies, judging whether can be used in the U.S organ transplant network. Now we discuss as following areas:

- Increasing the number of potential donors

The U.K has an ‘opt-in’ approach to organ donation. This requires potential donors and/or relatives to indicate their willingness for organs to be removed for transplantation. Individuals can indicate their wishes in a number of ways – by making their wishes known to relatives and those closest to them, by registering on the National Health Service (NHS) Organ Donor Register (ODR), or by carrying a donor card. Approximately 11.6 million people (19% of the U.K population) are registered on the NHS ODR. So the number of potential donors can increase rapidly through this approach.

- Increasing the number of living donors

Here is the figure of the Number of organ donors, transplants and patients on the end of year waiting lists in the U.K:

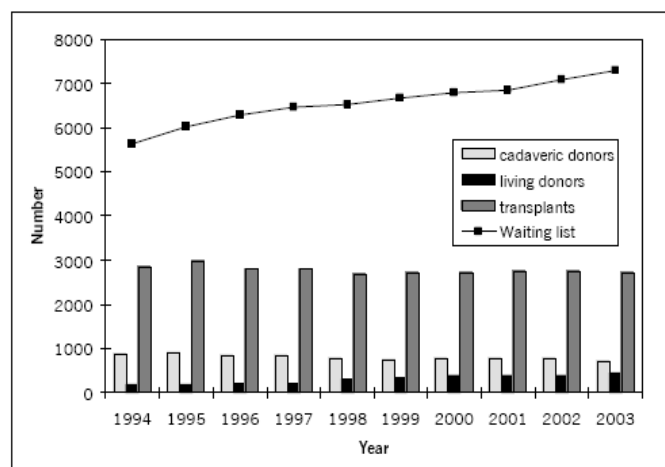


Figure 6: The annual data in U.K

According to the above figure, we can find that the quantity of living donors in U.K is increasing at a rapid speed. This is related to the U.K organ transplant policy closely. Here is some influential policies.^[9]

1. Authorities and Health Professionals should increase public awareness of the option for donating a kidney to a family member and the resulting benefits.
2. All nephrologists involved in caring for end-stage renal disease (ESRD) patients should be aware of the need to explore the donor option early with the family when a patient presents with early-stage renal disease.
3. ‘Living donor co-ordinators’ should be appointed to transplant units to integrate and oversee the exhaustive process of donor selection and health checks within a family.
4. Living related and unrelated donation should be encouraged, within national laws.

- The involvement of the government ——Changing legislation and Setting goals

Opinions on whether there should be a change in legislation differ. The British Medical Association, The British Transplantation Society (BTS), the Royal College of Surgeons and some patient groups, such as the National Kidney Research Fund (NKRF), consider the U.K should adopt a system of presumed consent with permission. They argue that although this system may make little difference to existing procedure it will affect the culture and framework within which discussions about donation take place, with the strong assumption being that the answer will be yes. NKRF has also suggested there should be a system of mandated choice which obliges everyone to consider whether they wish to donate their organs.

The Government announced that a new national Blood and Transplant Authority will replace U.K Transplant (UKT) and the National Blood Authority (NBA). UKT, which was established only in 2000, is widely regarded as a success, having maintained organ donation rates and increased the number of people on the ODR.

- Other policies

- 1) At 2005, U.K has introduced a new policy that giving a right to the doctor to maintain the vitality of the patients' some organs after their death. Then the doctor could persuade their families to donate the organ of the decedent later.
- 2) U.K government strongly supports the research in the field of organ transplant, including Clone technology, transgenic technology and so on.

Korea

Korea is also one of the countries with the developed technology of the organ transplant. An idea that originated in Korea is that of a kidney exchange system, which can take place either with a living donor or with the cadaver queue. In Korea, there are many styles of the donation. For example, paired-kidney donation, list paired donation, 3-way, 4-way or n-paired. When confronted with different situations, different methods can be applied to solve the problem.

Modifying the model in Task 1

According to the evaluating the OPTN policy at U.S and analyzing the organ transplant policies in other countries, we modify the model which has been designed in the Task 1. We improve the OPTN policy through adopting the better policies from other countries. The modification of the model just means the optimization the organ transplant network. So we can optimize each of the three parts of the network——Input, Organ Bank(OB),Output. See the specific measures in our one-page report which writing to the Congress.

One-page Report

Along with the continuing and rapid advances in medicine and health technology, the problem of the Orange Transplant is becoming a hot spot now. Meanwhile, the demand for organs for transplantation exceeds the number of donors. To help this situation, US Congress established the OPTN to match organ donors to patients with organ needs. However, no significant disparities for organ transplant rates for the sickest patients exist on a regional, gender or racial basis under the current OPTN policy, but the needs of the most critically ill patients are not met. Further more, there are a few disadvantages and limitations in the OPTN policy.

And then we have studied the organ transplant policies in other countries. Although the relevant policies and statutes in these countries are less comprehensive than that in U.S, there still a lot what U.S could learn from. Especially in the developed countries which own a comprehensive health organization, such as the U.K, France, Spain, Japan, Korea and so on. According to the analyzing and studying of the organ transplant policies in other countries, we optimize the current Organ Transplant Network in the U.S and improve the OPTN policy in order to adapt the actual organ transplant status. We optimize each of the three parts of the network——Input, Organ Bank(OB),Output.

For Input: In this part, we can achieve the optimization by consulting the reasonable policies in U.K, Spain and other countries. Our recommendations as follow:

- Establishing more Transplant Centre, strengthening the communion of the information among small networks;
- The nation amends or promulgates the relevant laws, the government sets detailed goals and calls on the people to be organ donors;
- Setting up a National Health Service Station as in U.K, registering the information of the person who wants to donate the organ, providing the especial services to the donors.

For OB: The core of the OB system is that the information will be placed into a waiting sequence and an available list, and a suitable matching algorithm should be found for the system. So in this part, we mainly improve the organ allocation strategy. Besides the point system allocation strategy which currently utilized by the UNOS, we also have compared three other policies: a first-come first-transplanted system; an efficiency-based algorithm that incorporated correlates of patient and graft survival; and a distributive efficiency algorithm. Compare with the point system allocation strategy, each algorithm has its own advantage. According the computer simulation, we find that the efficiency-based organ allocation strategies in cadaver kidney transplantation would yield improved equity and efficiency measures compared with existing algorithms. So we can combine the point system allocation with the efficiency-based algorithm, making the match more fair and effective.

For Output: In this part, it needs to improve the techniques of the organ transplant constantly, to avoid the death of the patients after the operation. So the government should increase the investment to carry on the technical studies, including Clone technology, transgenic technology and so on.

Finally, we also hold a proposal of establishing a market for orange exchange. At this market, people can do the business of the organs. However, the business can not violate the criteria and constraints of the ethics, politics and many other aspects.

Task 3

Problem analysis

Recently, there has started to be kidney exchanges involving two donor-patient pairs such that each donor cannot give a kidney to the intended recipient because of immunological incompatibility, but each patient can receive a kidney from the other donor. Exchanges are also made in which a donor-patient pair makes a donation to someone on the queue for a cadaver kidney, in return for the patient in the pair receiving the highest priority for a compatible cadaver kidney when one becomes available. We explore how such exchanges can be arranged efficiently and incentive compatibly.

In this section, our aim is to devise a procedure to maximize the number and quality of exchanges, taking into account the medical and psychological dynamics of the situation. Consider the n patients with a non-compatible living donor, we can extend the paired exchange to a more complex exchange model and build a kidney exchange mechanism Top Trading Cycles and Chains (*TTCC*), a systematic procedure to select a matching for each kidney exchange problem. In our paper we can introduce a Rational-Cautious ($R-C$) preference mechanism and then build a *TTCC-RC* Model for some improvement.

Patients who have no living donors, but are on the queue for cadaver kidneys, are not explicitly included as agents in the kidney exchange problem as we discuss it here. The algorithm of the kidneys exchange is mainly based on a reference paper ^[12]. The preferences of patients over available kidneys can be defined by involving the age, HLA match degree and other factors. And this preference has been improved in a $R-C$ mechanism. According to a certain chain selection rule and the preference function defined, a simulation can be obtained in the model. The chain selection rule and the preference function can be modified in a degree to maximize the number and quality of exchanges in our model.

Definition and notation

As it is analyzed above, we denote a donor-recipient pair by (k_i, t_i) , where k_i is noted as the available kidney i , and t_i as the patient i . Let K denote the set of living donor kidneys at a specific point in time. While patients and their doctors may define their preferences over possible kidneys as they wish. Given any patient t , part of K is outside the consumption set due to ABO blood-type incompatibility or a positive cross-match. For each patient t_i , let $K_i \subset K$ denote the set of living donor kidneys that are compatible with patient t_i . Note that kidney k_i may or may not be in K_i .

Let ω denote the option of entering the waiting list with priority rejecting the donation of his donor's kidney k_i , and P_i denote his strict preferences over $K_i \cup \{k_i, \omega\}$. For our purposes the relevant part of P_i is the ranking up to kidney k_i or ω , whichever ranks higher.

If patient t_i ranks kidney k_i at the top of his preferences that means he does not wish to participate in an exchange. If patient t_i ranks k_i on top of ω that means he (or his donor) does not consider exchanging kidney k_i with a priority in the cadaver kidney waiting list.

We are finally ready to formalize a (static) kidney exchange problem. A kidney exchange problem consists of:

- 1) A set of donor-recipient pairs $\{(k_1, t_1), \dots, (k_n, t_n)\}$;
- 2) A set of compatible kidneys $K_i \subset K = \{k_1, \dots, k_n\}$ for each patient t_i ;
- 3) A strict preference relation P_i over $K_i \cup \{k_i, \omega\}$. for each patient t_i .

Let R_i denote the weak preference relation associated with the strict preference relation P_i .

Note that, since the preferences are strict, given two kidneys k, k_0 if $k R_i k_0$ and $k \neq k_0$ then $k P_i k_0$.

The outcome of a kidney exchange problem is a matching μ of kidneys/waitlist option to patients such that:

- 1) Each patient t_i is either assigned a kidney in $K_i \cup \{k_i\}$ or the waitlist option ω ,
- 2) No kidney can be assigned to more than one patient although the waitlist option ω can be assigned to several patients. (This latter possibility makes the kidney exchange problem different from the housing problems formulated above.)

Let $\mu(t_i)$ denote the assignment of patient t_i under matching μ .

Patients who have no living donors, but are on the queue for cadaver kidneys, are not explicitly included as agents in the kidney exchange problem as we model it here, so it is with the priority rules modeling by which the cadaver queue is managed.

Model design——TTCC-RC Model

A kidney exchange mechanism is a systematic procedure to select a matching for each kidney exchange problem. We are almost ready to introduce the *TTCC* mechanism, a variant of *TTC* mechanisms, for kidney exchange. Before we give a few definitions and observations to facilitate the description of the mechanism, and a *R-C* preference mechanism add on this issue to a certain extent.

Cycles and w-Chains

As it is presented in Kidney Exchange ^[12], the mechanism relies on an algorithm

consisting of several rounds (no more than the number of the agents). It is easy to see that w-chains are associated with indirect exchanges. The notion of a maximal w-chain does not resolve the potential conflict between w-chains; a kidney or a patient may even be part of several maximal w-chains. Therefore the selection process of w-chains in the algorithm is not as straightforward as cycles.

A lemma is given in this part as follows:

Consider a graph in which both the patient and the kidney of each pair are distinct nodes as is the waitlist option w . Suppose each patient points either towards a kidney or w , and each kidney points to its paired recipient. Then either there exists a cycle or each pair initiates a w-chain.

This lemma has been proved by using graph theory terminology in ^[12].

In each round, each patient t_i points either towards a kidney in $K_i \cup \{k_i\}$ or towards w , and each kidney k_i points to its paired recipient t_i .

A cycle is an ordered list of kidneys and patients $(k_{01}, t_{01}, k_{02}, t_{02}, \dots, k_{0m}, t_{0m})$ such that kidney k'_1 points to patient t'_1 , and then patient t'_1 points to kidney k'_2

For a simple example as follows:

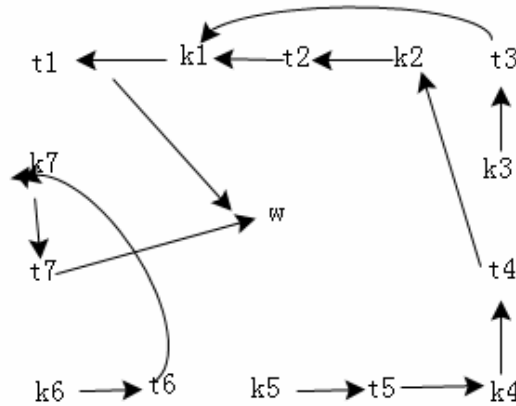


Figure 7: Kidney exchange and match

In this graph, each kidney-patient pair initiates (i.e. is the tail of) a w-chain:

$W_1 = (k_1, t_1), \quad W_2 = (k_2, t_2, k_1, t_1), \quad W_3 = (k_3, t_3, k_1, t_1), \quad W_4 = (k_4, t_4, k_2, t_2, k_1, t_1),$
 $W_5 = (k_5, t_5, k_4, t_4, k_2, t_2, k_1, t_1), \quad W_6 = (k_6, t_6, k_7, t_7), \quad W_7 = (k_7, t_7).$

The Exchange Mechanism

The key issue of the dynamics of the *TTCC* mechanism is to find an appropriate chain selection rule to when one among multiple chains must be selected. Once the rule is defined, the algorithm can also be identified for the problem. It has been discussed in detail in Kidney Exchange ^[12].

Knowledge of graph theory, and set theory are involved in this part. At the end of the

procedure, each patient with a living donor is assigned a kidney (or a high priority place on the waiting list).

It is worth emphasizing that the chain selection policy does not affect a patient who is at the head of a chain: Since he points to the waitlist option, he will eventually be selected regardless of the chain selection rule. However whether his intended donor's kidney is offered to the cadaver waiting list or another patient with a living donor depends on the rule.

Each potential donor is an agent with his own incentives. Clearly it will cause serious complications if a donor withdraws his consent after his intended recipient is transplanted a kidney through an exchange arrangement. A similar practice should be adopted to carry out the exchanges under *TTCC*, and, following the final assignment of kidneys, all transplantations associated with a given cycle of the *TTCC* algorithm that produced the assignment should be performed simultaneously. Practice needs to be adopted to perform the transplantations associated with w-chains that are selected and (eventually) removed by the algorithm. There is one last constraint on the timing of transplantations. Whenever the transplantations associated with a cycle/w-chain fail to be performed due to a last-minute difficulty, this not only affects the patients in this cycle/w-chain but potentially other patients whose assignments were finalized by the algorithm in the subsequent rounds.

Feasible Chain Selection Rules

For the defined mechanism, we assume that when one among multiple chains must be selected, a fixed chain selection rule is invoked to make the choice. We will consider a number of such chain selection rules, and their implications for incentives, efficiency, and equity.

Depending on policy priorities, one may consider adopting a number of alternative chain selection rules. So many specific chain selection rules have been given out in Kidney Exchange ^[12]. And the efficiency of these rules has been proved. We present some classical and practical rules as follows:

- 1) the rule that chooses the longest w-chain and keeps it;
- 2) Choose minimal w-chains and remove them;
- 3) Prioritize patient-donor pairs in a single list. Choose the w-chain starting with the highest priority pair and remove it;
- 4) Prioritize patient-donor pairs in a single list. Choose the w-chain starting with the highest priority pair and keep it.

Take efficiency and incentives into consideration, that is to maximize the number of exchanges in some extend, we discuss in what way these rules make our procedure achieve a maximum.

Among these three chain selection rules, the forth rule yields an efficient and strategy-proof mechanism. A chain selection rule which chooses among longest w-chains upsets strategy proofness, because it allows a patient to influence his assignment by influencing the lengths of w-chains via a preference misrepresentation.

In each round of the choice among multiple chains, this rule takes the longest chain in order to increase the total exchange number. In this way, this rule can be used to ensure as many as possible of the kidney exchanges. Then in our model we take use of the last one.

R-C Preference Construction

Preference construction are discussed rationally and cautiously in a separate way in ^[12], the preferences of patients over available kidneys (or equivalently living-donors) in the sample are determined using the survival analysis of grafts reported in the Mandal et al. And here we can make some improvements in the preference construction issue.

Preference construction are introduced to describe the matching degree of the patient, a stronger preference means a higher priority of the matching exchange and it can lead to a better quality of exchanges. Here in our model, we take the medical and psychological dynamics of the situation into account and innovatively construct a R-C Preference Mechanism (the combination of rational and cautious case) to maximize the exchange quality.

We assume that the utility function of each patient depends on the donor age and the number of HLA mismatches throughout our simulations, Mandal et al. Several factors such as patient race, patient age, patient health condition (more specifically previous history of diabetes), patient gender, donor type (cadaver or live), donor age, HLA mismatches can be involved from the medical perspective.

Using the above analysis, and assuming that a patient's utility from a graft is a monotone decreasing function of its relative risk of failure determined by the age of the living-donor and the HLA mismatch number, Therefore, since a patient's utility over grafts is a monotone decreasing function of its relative risk of failure, the utility function as follow:

$$u(x, y) = \omega_1 \cdot \frac{x^* - x}{x^*} + \omega_2 \cdot \frac{y^* - y}{y^*};$$

Where: x denote the HLA mismatch number between the patient t_i and his compatible donors, and x^* the average level (changes in the range $\{0, 1, 2, \dots, 6\}$), y is the age of the living-donor (changing in the open interval $(18, 80)$).

ω_1 and ω_2 are the weight of the two indexes, they can be defined by the public view and medical theory.

- When $u(x, y) > 0$, we consider that the exchange is very acceptable;
- When $u(x, y) < 0$, we take the psychological changes of the patient into consideration, that is the patient may reject to the exchange program. This possibility may be different for different patients, some may have urgency for a kidney transplant or some others may be particularly cautious. We just give out the statistical probability. This assumption may be realistic especially when patients do not know much about the previous health conditions of other donors. Once the patient t_i reject to the exchange program, the exchange should be run another time, with the patient t_i and his living donor removed.

Using this utility function, we construct patient preferences over compatible donors. Under the integrated $R-C$ preference scenario, we can undertake a proper chain selection rule and then achieve a large number of exchanges and a relative better quality.

Simulations example

Statistic data information

We obtain the data from the OPTN Website ^[2] and data banks ^[3].

And some data processing we can get the relevant statistical data information of patient and donor characteristics for the simulation.

Table 5: Age distributions for new patients used in the simulations

	Deceased Donor	Percent	Living Donor	Percent	All Donor Types	Percent
Pediatric	17,629	18.02%	48	0.06%	17,677	9.93%
Adult	80,139	81.92%	80,049	99.91%	160,188	90.02%
Unknown	52	0.05%	24	0.03%	76	0.04%
All Ages	97,820	1	80,121	1	177,941	1

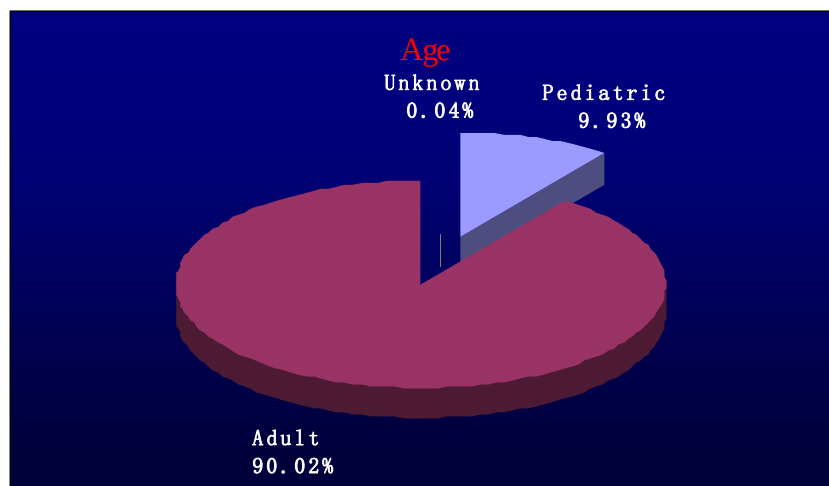
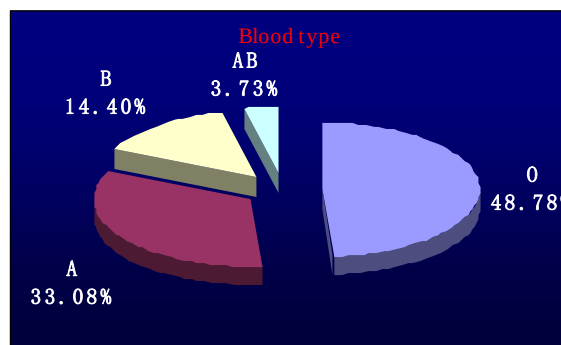


Figure 8: The percent of each age

Table 6 (figure 9): ABO Blood-type distributions for new patients used in the simulations

Blood type	Percent
O	48.78%
A	33.08%
B	14.40%
AB	3.73%



This data information can be used to control the selection of random numbers in the simulation process.

Three simulations and Results

In our simulations, we randomly simulate a sample of n donor-patient pairs using the population characteristics explained above. Then, we determine the preferences of patients over kidneys in the sample: For each patient t_i , we first check whether a donor k_j is ABO-compatible. If k_j is ABO-compatible, then we check whether there is a positive cross-match between t_i and k_j . If they test negative for cross-match, then k_j is in the compatible donor set K_i of patient t_i . After finding the set of compatible kidneys for each patient, we obtain a preference ordering on this set, using the utility functions described above. We construct a set of preferences for each patient using the $R-C$ preference construction methods discussed above and assuming that 30% patients with incompatible donors consider the waitlist option or the Low Quality ($u(x, y) < 0$) acceptable.

We simulate each of three mechanisms under the preference scenario, including the No-Exchange, Paired/direct and *TTCC* mechanisms. We use a Monte-Carlo Simulation (See the specific procedures in Appendix II) size of 100 trials for three different population sizes of 30, 100, and 300 pairs. Then a statistical simulation result can be obtained as follows:

Table 7: The R-C preference simulation result (The first simulation)

Mechanism	Total-Trans%	Own-Donor-Trans%	Trade%	Waitlist-Up-grade%	Risk	HLA-mis
No-Exchange	39.64	39.64	0	0	1.93	3.02
Paired/direct	59.12	39.64	19.48	0	1.93	3
TTCC	84.75	15.72	69.03	15.25	0.87	1.94

($n=400$, Low quality exchange or waiting list =30%)

Table 8: The R-C preference simulation result (The second simulation)

Mechanism	Total-Trans%	Own-Donor-Trans%	Trade%	Waitlist-Up-grade%	Risk	HLA-mis
No-Exchange	38.15	38.15	0	0	1.93	3.03
Paired/direct	58.34	38.15	20.19	0	1.93	3.02
TTCC	82.27	14.81	67.46	17.73	0.91	2.03

($n=200$, Low quality exchange or waiting list =30%)

Table 7 reports the general patient statistics under each regime in the columns. The first column in these tables reports the total live donor transplants as percentage of the population size, which is the sum of next two columns, transplants from own compatible donor and transplants from trades. The forth column is the percentage of patients upgraded to the top of the waitlist as heads of w -chains. The fifth and sixth columns report the quality of matches in the live donor transplants: the risk of graft failure relative to the risk under no-exchange

mechanism with population size $n=400$ is reported in the fifth column and the number of HLA mismatches for an average transplant is reported in the sixth column. In the table 8, we change the n into 200. By comparison, we can found that the matching proportion become little and the matching quality will get worse as the total number of the patients decrease. The result is consistent with the reality. The 30% probability of the waiting list or low quality exchange is an adjustable parameter.

Form the table, we find that Paired/direct exchange can optimize the quality and efficiency of the kidney transplant Matching. And the $TTCC$ under the assembled $R-C$ preference utility model ($TTCC-RC$) performs very the best, taking the medical and psychological dynamics of the situation into consideration.

Conclusion

As analysis in the chain selection rules part, the rule we adopt is to prioritize patient-donor pairs in a single list. Choose the w-chain starting with the highest priority pair and keep it. This yields an efficient and strategy-proof mechanism. A chain selection rule which chooses among longest w-chains upsets strategy proofness, because it allows a patient to influence his assignment by influencing the lengths of w-chains via a preference misrepresentation. In each round of the choice among multiple chains, this rule takes the longest chain in order to increase the total exchange number. In this way, this rule can be used to ensure as many as possible of the kidney exchanges.

The $R-C$ preference construction is a quite enjoyable way to take both medical and psychological dynamics of the situation into consideration, when we are aimed to reach a good quality of exchange.

From the simulation result, we can see the Total-Trans% and Waitlist-Up-grade% have been greatly improved after the use of the $TTCC-RC$ exchange model.

Once the kidney exchange program is used, the waiting list became two separate lists, namely the list with a living donor for exchange and the cadaver queue. These two lists have some mutual effect on each other, higher priority should be distributed to candidates who come to the cadaver queue from the kidney exchange mechanism and whose donor makes a donation to someone on the queue. And then the cadaver list should be arranged over again. Sometimes an exchange between the two lists is possible.

The comprehensive kidney exchange mechanism in our model will potential benefits:

- Allow not only for paired-kidney-exchange and indirect exchange but also many other forms of exchange such as a three-way exchange, more elaborate forms of indirect exchange, etc.;
- Benefit not only the incompatible pairs but also the compatible ones,
- Eliminate or at least reduce the adverse affect to O blood-type patients with no living donors;
- Organize the live kidney exchange in a way that is compatible with the existing system of cadaver kidney allocation.

This $TTCC-RC$ model can be extended to kidney exchange with multiple donors.

Task 4

Problem analysis

Patients may face agonizing choices. For example, suppose a barely compatible, in terms of HLA mismatch kidney becomes available from the cadaver queue. Should they take it or wait for a better match from the cadaver queue or from an exchange? In particular, a cadaver kidney has a shorter half-life than a live donor kidney. From the patients' perspective, in order to achieve a better transplant, a rejection to the offer kidney is allowed. However, the corresponding risk grows and it is hard for the patient to do his choice.

Then the key issue of this problem in our study is to consider the risks, alternatives, and probabilities in our analysis. Firstly, we can build a dynamic programming model in the continuous case, and then extend it into the discrete case for a Numerical solution through the iterative method. A Quality Evaluation System should be involved to define the health level of the patient and the kidney quality.

For those with an available donor and have a choice to participate in a kidney exchange, if they obtain a more acceptable kidney, we suppose they will take the new offered kidney. Otherwise, if they give out their owned donor but without receiving a new one, it is with the case of the general rejection. And here we can give him a higher priority on the waiting list.

Model design—Decision Model under dynamic programming

Definition and notation

We denote q as the kidney quality $q \in (0, Q)$, as for a patient t_i in health state $h \in (0, H]$. If the patient t_i rejects the available organ, the patient's health evolves according to a first-order Markov process described by $f(h' | h)$. Assume that $f(h' | h_1)$ weakly first-order stochastically dominates $f(h' | h_2)$ for $h_1 > h_2$. The state $h = 0$ corresponds to death. The patient will receive a draw from the kidney quality distribution $f(q)$ each period until he accepts a kidney or the patient dies. The state $q = 0$ describes a period in which the patient does not receive a kidney offer (and in reality is the state that occurs with the highest probability). If the patient t_i accepts an offered kidney for transplant in period t , the probability that the operation is a success in period $t+1$ as a function of the patient's period t health and period t kidney is $p(h, q)$ where $p_h(h, q) > 0$, $p_q(h, q) > 0$, $p(0, q) = 0$, and $p(h, 0) = 0$.

• The Continuous case ^[15]

The function of the probability that the operation is a success is a simplistic representation of the impact of kidney quality and patient health on the expected benefit from transplantation. A more realistic model would specify a complete survival curve as a function of q and h , both q and h are supposed to be continuous here.

A successful transplant confers an immediate benefit of $B(q, h)$ to the patient while a

failed transplant results in immediate death, with a payoff of 0 (we ignore here the possibility that a patient whose graft fails may be placed back on the waiting list to await retransplantation). The benefit of $B(q, h)$ can also be defined in a simple way. Assume that the patient's single period utility when alive is u (including the period in which a patient's physician accepts a kidney), and let the discount factor be δ . The expected $t + 1$ payoff from accepting a kidney in t is

$$EV^{TX}(h, q) = p(h, q) * B(h, q) \quad (1)$$

$EV^{TX}(h, q)$ is an average level of the payoff, and $B(q, h)$ can be defined to be a constant.

The expected payoff from rejecting a kidney and continuing to wait is

$$EV^W(h) = \int_q \int_h V^W(h', q') f(h|h') f(q') dh' dq' \quad (2)$$

Where $V^W(h, q)$ is defined by the fixed point of the Bellman equation.

$$V^W(h, q) = u + \delta \times \max\{EV^{TX}(h, q), EV^W(h)\} \quad (3)$$

The patient should accept the available kidney if $EV^{TX}(h, q) > EV^W(h)$. The set of h 's and q 's satisfying this condition is depicted as region A (for "accept") in Fig. 1. The set of h 's and q 's satisfying $EV^{TX}(h, q) < EV^W(h)$ is depicted as region R (for "reject"). The organ quality cutoff, q^c for a given level of health is defined by the implicit function:

$$EV^{TX}(h, q^c) = EV^W(h) \quad (4)$$

The h, q pairs satisfying 4 comprise the set $A \cap R$. From 4,

$$\frac{dq^c}{dh} = - \frac{EV^{TX} - EV^W}{EV_{qc}^{TX}} \quad (5)$$

The sign of 5 depends on the relative magnitudes of $p_h(h, q)$ and $f_h(h' | h)$. Thus, it is an empirical question whether dq^c / dh is positive or negative. Fig. 1 depicts the case where $dq^c / dh > 0$.

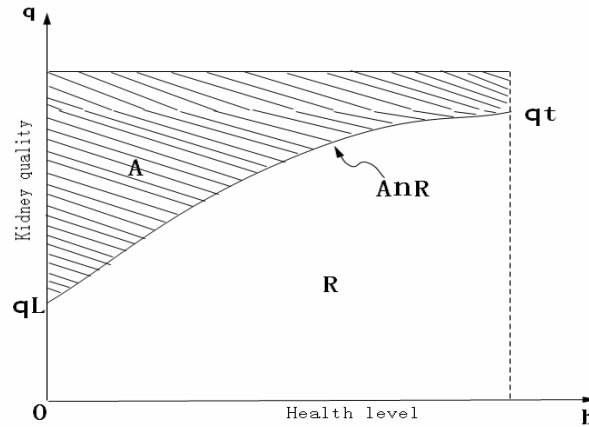


Figure10: The accept/reject rule.

The decision interval is very specific by the figure above, the shadow region stand for the acceptable part. The Curve upward trend tells us that the patient with a higher health level will mostly ask for a better quality kidney. And the points q_l and q_t show that when the health level of the patient is too high (near to the healthy status) or too low (near to the death status), a kidney with very good or low quality becomes unnecessary for him.

In this part, the dynamic programming is mainly based on the Bellman equation, where u is the Performance indicators of this optimal control process. The continuous solution to this equation may result in an image result (the patient's decision), however, the analytical solution is very difficult to find.

● The discrete case

The decision model designed in the continuous case sound very good, though it is hard to get the analytical answer. So we can convert it into the discrete case that is to consider the kidney quantity q and the level of health h of the patient as discrete variables. According to the health of the patients and the quality of the donor's kidney, we senate some rules to create a certain integer variables for p and h , then the equation 2 can be modified as follows:

$$EV^W(h) = \sum_q \sum_h V^W(h', q') * f(h|h') * f(q')$$

Variable definitions and other identity statements are the same with the continuous case. And then it can be defined as follows:

- Dynamic planning objectives: Maximize μ
- State variables: μ_t, h_t, q_t
- Control variables: $x_t = 0, 1$ when $q_t = 0$, x_t must be 0
- State transfer equation:
$$\mu_{t+1} = \begin{cases} u_t + \delta \times EV^{TX}(h, q) & x_t = 1 \\ u_t + \delta \times EV^W(h) & x_t = 0 \end{cases}$$

The iterative algorithm can be used here to solve this dynamic planning problem. For a given threshold ε , we can find a relative optimum solution to this problem. And the specific solution to this problem will be stated in the solution part (C++ Programming).

The evaluation of q and h and the Markov model

For a given patient, the level of his health h and his donor's kidney quality q should be quantified. The detailed evaluation of mathematical model is necessary, the evaluation of the health level and the kidney quality q may involve many metical factors such as the patient age, gender, some test indicators, blood pressure and temperature, miscellaneous diseases relevant to the health level in a long run and so on. The factors have influence on the donor donor's kidney quality q may also be various, including age, gender, etc.

When we consider the patient's health level, we collect the health information, and then

define the level status by reference of the OPTN policy and the medical knowledge, we make every attempt to group patients in different years into comparable statuses, and here we simply use the fuzz math theory to get a patient's health level. Different indicators may have different weight in the valuation progress. The value of h may change from 1 to 5; where a higher h means a better health condition.

In reality, the change of patient's health level h is a first-level Markov process^[16]. That is h at the time t can only be affected by the h at time $t-1$. So we can use a transition probability matrix to describe the transition situation of the patient's health level.

This matrix can be defined after a statistical process has been done in a large sample of the kidney patient. Here we use a transition probability figure to describe the problem.

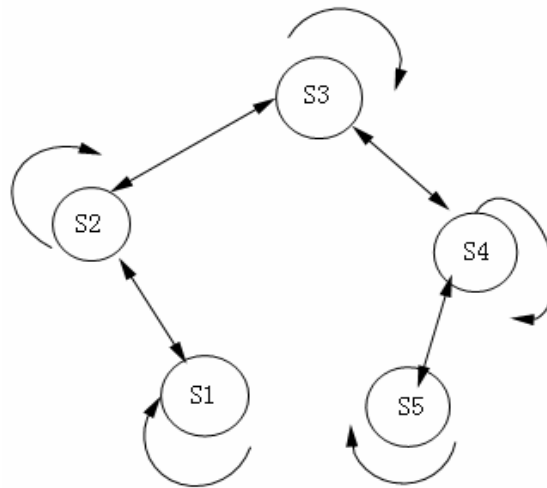


Figure 11: Markov Process graph

As for the evaluation of the donor's kidney quality may be almost the same, only with some different indicators and different weights of these indicators.

Because the measures of organ quality are highly correlated and looking at each individually does not provide a comprehensive picture, I create and compare across health levels a summary measure of organ quality. To construct the measure, I select a 20% random sample of the data and regress donor characteristics Q and patient controls X on a dummy y equal to one if the patient's graft failed within the first 3 months following transplantation: $y = X\beta_1 + Q\beta_2 + \varepsilon$. The weight can be modified in a degree to obtain a better result.

Since the quality of the donor's kidney q is at random, we can also obtain its distribution in a statistical way. In our model, we can assume that the q is a given distribution as the statistical method is not realistic. Once the special distribution is defined, a decision mechanism based on the dynamic programming is available.

Solution and result

● Empirical estimates

The most direct approach to determine the sign of δ would make use of observations on actual acceptances and rejections. A simple method used in ^[15] should be discussed at first.

The statistical data covering the kidney patients can be obtained after a large sample survey. Formally, these data should consist of observations on h and q for $\{h, q\} \in A$. Using these data, however, one can infer the sign of δ from the correlation between h and q . The expected organ quality among patients transplanted in a given level of health is $E(q|h > q^c)$, and it is straightforward to show that $d/dhE(q|h > q^c) > 0$ implies $dq^c/dh > 0$. In other words, if the line in Fig. 1 separating regions A and R is truly upward sloping, there should be a positive correlation between h and q among transplanted patients. It is also assumed that distribution of organ draws received by a patient is independent of the patient's health. That is $f(q')$ is not conditional on h . this assumption is clearly consistent with the realistic.

This method is based on the empirical estimates, it is easy to make a proper decision by Statistical analysis of data and some relevant decision help factors.

● Iterative algorithm

From the definition part, we suppose that a patient is with a initial status h_0 and μ_0 , and the kidney quality distribution $f(q)$ at each period is the same. We assume that the kidney quality holds the following distribution.

$$\begin{pmatrix} q & 0 & 1 & 2 & 3 & 4 & 5 \\ p(q) & 0.90 & 0.02 & 0.02 & 0.02 & 0.02 & 0.02 \end{pmatrix}$$

Then the Markov process transition probability matrix should be defined as follows:

$$P(h'|h) = \begin{bmatrix} & 1 & 2 & 3 & 4 & 5 \\ 1 & 1/2 & 1/2 & 0 & 0 & 0 \\ 2 & 1/3 & 1/3 & 1/3 & 0 & 0 \\ 3 & 0 & 1/3 & 1/3 & 1/3 & 0 \\ 4 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 5 & 0 & 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

The translation happens only between the neighboring states.

The iterative algorithm can be described as follows:

$$EV^{TX}(h, q^c) = EV^W(h)$$

$$EV^W(h) = \sum_q \sum_h V^W(h', q') \bullet f(h|h') \bullet f(q') ;$$

$$Threshold = 1E-8;$$

$$While(u_{t+1} - u_t > Threshold)$$

$$\{ \quad u_t = u_{t+1}$$

$$If(x_t == 1)$$

$$\mu_{t+1} = u_t + \delta \times EV^{TX}(h, q) ;$$

$$Else$$

$$\mu_{t+1} = u_t + \delta \times EV^W(h);$$

$$\} \quad // \text{ through a threshold to control the number of iterations.}$$

In the reality, the relevant probability should be modified after a statistical process of a large sample upon patients. For example the kidney quality distribution $f(q)$ can be modified better to help the patient to make the decision.

Exchange choice

Another choice the patient can make is to participate in a kidney exchange. In our model, we have not taken this choice in consideration. Here we can do some work to modify the model to this special case.

For those with an available donor and have a choice to participate in a kidney exchange, if they obtain a more acceptable kidney, we suppose they will take the new offered kidney. Otherwise, if they give out their owned donor but without receiving a new one, it is with the case of the general rejection. And here we can give him a higher priority on the waiting list.

Let the patient p with an existing state (h, q) , once the (h, q) was located in the acceptable region, we suppose that he will accept the existing available kidney. While the (h, q) is located in the reject region, then he will have two choice, exchange the available kidney or reject it. The exchange choice is based on a better situation. So if the situation becomes better then it may convert into the problem in the above model. Otherwise, give him a higher priority and then the $f(q)$ can be modified especially for the patient p .

Task 5

Analysis on Ethical and Political issues

Organ transplantation has been recognized as one of the most gripping medical advances of the century as it provides a way of giving the gift of life to patients with terminal failure of vital organs, which requires the participation of other fellow human beings and of society by donating organs from deceased or living individuals. The increasing incidence of vital organ failure and the inadequate supply of organs, especially from cadavers, having created a wide gap between organ supply and organ demand, which has resulted in very long waiting times to receive an organ as well as an increasing number of deaths while waiting. These events have raised many ethical, moral, societal, cultural and political issues regarding supply, the methods of organ allocation. It has also led to the practice of organ sale by entrepreneurs for financial gains in some parts the world through exploitation of the poor, for the benefit of the wealthy. Now we focus on the ethical and political issues to study the organ transplantation.

Ethical issues ^[9]

Due to the serious shortage of donors, the analysis on the ethical problem about organ transplantation had been taking more attention. An indispensable step of the organ transplantation is that the useful organ must be transplanted from the donor to the patient. So whether is this act ethical? This is the first ethical problem which has been arguing intensely. Here are some basic ethical principles.

1) Primary ethical principles

A number of primary principles are widely accepted as forming the bedrock of medical ethics. In individual cases, conflict often arises in trying to adhere to all of these principles at the same time.

- **First principle**

Voluntary contribution is the first ethical principle in the source of the donor. Medical principles and social standards should be taken into account on the problem of acceptor selection.

- **Beneficence: Doing good**

A central tenet of medical ethics is the obligation to strive at all times to do good for the patient. Although no physical good will accrue to a donor it is generally accepted that the psycho-social benefits to the living donor justify the risks involved.

- **Non-maleficence: Avoiding harm**

Making sure that the balance between benefit and harm is appropriate is an important clinical judgment. A high standard of donor assessment and risk limitation is therefore of paramount importance before living kidney donation takes place.

- **Respect for autonomy**

Patients with the capacity to understand relevant information, to consider its implications, and to come to a communicable decision are deemed to have decision-making capacity. Their decision to donate should therefore be respected.

- **Justice: Promoting fairness**

The principle of justice is very important in kidney distribution, where demand far outstrips supply. In that context, the allocation of organs requires a ranking system in order of priority, with moral justification for the method by which a patient's ranking is decided. In transplantation, scarce resources usually have to be carefully allocated to recipients chosen from a larger pool of the population.

2) Cadaver organ donation

There has been an increase in living-donor organ procurement in recent years. The bulk of available organs still come from cadaver donors, brain-dead donors, along with organs from the Non-heart-beating donor (NHBD) procurement program now used by a number of transplant centers. However, this resource base is shrinking. This, in combination with the ever-increasing rise in potential recipients, is a cause of considerable pressure on transplantation program.

- **Cadaver organ donor**

In most countries, obtaining consent to proceed with organ donation is a major barrier to overcome. The process of gaining formal consent from relatives or from the patient during life can be defined as 'opting in' to a donor scheme. Unless consent is expressly given, the presumption is that consent is withheld. In a number of European countries, the opposite pertains. Consent is presumed unless the patient has specifically opted out before death. This type of legislation appears to influence organ donation favorably, but may be seen as intruding on the individual's rights.

- **Allocation of cadaver organs**

Who 'owns' cadaver organs and who makes the decision regarding allocation are both issues in need of clarification. However, there is a general presumption that the State holds the responsibility for allocation or disposal of the organs, which it discharges by delegation to the appropriate transplant team. To date, the notion that cadaver donation and allocation can be made conditional upon the personal attributes. All such kidney distribution systems should be transparent and regularly audited.

- **Brain death principle**

It was proposed that implementation of brain-death legislation, setting up the newly ethical conception and prohibiting the organ business.

3) Living organ donors

The ethical approach to organ donation is conditioned principally by those rules that seek to be charitable while preserving autonomy. Given the increasing success of living donor transplant, as judged by graft and patient survival, and given the scarcity of cadaver organs, living-donor transplant has hitherto been regarded as a regrettable necessity; but the chronic shortage of cadaver organs as recently caused a more general acceptance of LD transplants. The results are better than cadaver grafts, the donors' autonomy is respected, and the act is recognized as emotionally fulfilling for donors as well as recipients.

- **Payment of altruistic donors**

The cornerstone of clinical transplantation has been the altruistic donation of kidneys from living relatives. Societies which support the development of

transplantation have generally refused to assign a monetary value to a transplantable organ or tissue: the gift of a transplant is therefore priceless, and legal control now exists in European countries to prevent payments for living related organs. The World Health Organization has stated that the body and its parts cannot be the subject of commercial transactions, and all giving and receiving of payments should be prohibited.

Recommendations on ethics

- It is the right of individuals to donate as well as to receive an organ;
- Commercially motivated renal transplantation is unacceptable, has been widely prohibited by law, and the International Society of Transplantation strongly opposes their practice;
- Given the increasing success of living donor transplants as judged by graft and patient survival, and given the scarcity of cadaver organs, living-donor transplants should be encouraged. The appeal for using living donors in renal transplantation is in part due to the continuing shortage of cadaver donors;
- The altruistic living donor must give informed consent, which can only be obtained if he has a proper understanding of the risk involved;
- A patient should be treated as an 'end' and not as a 'means'. Respect for dignity, integrity and authenticity of the person is a basic human right;
- Acceptance of living unrelated donors should be done only after the local ethical committee has given permission or, as required by the country, permission by the Courts;
- In the last instance, what is and what is not ethical should be determined by the balance between clinical utilitarian demand (saving lives in a cost-effective way) and respect for an individual's right to donate or not to donate an organ in life or after death.

Political issues

Among the various factors, the policy issue plays a most significant impact on the organ transplant. The nation can establish the national organ transplant network by formulating the policies. The government can change the relevant laws and regulations, set goals, increase the investment and so on. The organ transplant network becomes comprehensive or even worth under these steps. e.g., recent laws have been passed in the U.S that forbid the selling or mandating the donation of the organs; At 2005, U.K has introduced a new policy that giving a right to the doctor to maintain the vitality of the patients' some organs after their death.

Recommendations on policy

- Organ donation is valuable and should be considered a charitable gift. Society should express gratitude to the organ donors for their gift as is done with other charitable contributions, without jeopardizing its altruistic basis (e.g., Medal of Honor, limited reimbursement, medical leave, priority access to organ for transplant, donor insurance);

- ‘Living donor co-ordinators’ should be appointed to transplant units to integrate and oversee the exhaustive process of donor selection and health checks within a family;
- Establishing more large centers in each region;
- Establishing educational programs that will improve patient care and quality of life.
- Acting immediately to provide adequate payments to assure that end-stage renal disease (ESRD) payments support high quality care.
- Professional organizations, within countries, should, where necessary, maintain pressure on Government Health Departments to maintain an adequate number of intensive care beds; to create a cadre of national transplant co-ordinators; to fund and deploy educational programs for intensive care physicians, such as EDHEP; and encourage initiatives such as Donor Action;
- Laparoscopic nephrectomy offers donors less post-operative morbidity, quicker convalescence and better cosmetic results. It increases the number of individuals willing to donate without increasing the risk to donors’ safety or allograft function. It should be used, whenever possible, only by appropriately trained and experienced surgeons;
- Paired kidney exchange if permitted by National Law, should be encouraged where appropriate donor/recipient pairs are available;
- Authorities and Health Professionals should increase public awareness of the option for donating a kidney to a family member and the resulting benefits;
- All nephrologists involved in caring for ESRD patients should be aware of the need to explore the donor option early with the family when a patient presents with early-stage renal disease.

Changing and Ranking order the criteria

The criteria we will use for priority and placement include:

1) Waiting time

As general, waiting time is an importance factor we will consider.

2) ABO compatible

ABO compatibility is a very important factor for kidney transplants. Virtually 100% of ABO incompatible kidneys will be rejected within minutes.

3) HLA mismatch

HLA matching is used to measure immune system compatibility between an organ donor and a patient. Those patients with a higher level of HLA match traditionally have received higher transplantation priority. Some minority patients, however, are less well matched with Caucasian donors because HLA patterns are inherited. This may result in a longer wait for minorities who need transplantation.

4) Age

Table 9: The annual number of the donors in different age

	2000	2001	2002	2003	2004	2005	2006	To Date
Pediatric	908	908	858	794	870	798	817	17, 677
Adult	10, 065	10, 657	11, 017	11, 431	12, 102	12, 470	11, 696	160, 188
Unknown	9	1	3	1	0	2	9	76
All Ages	10, 982	11, 566	11, 878	12, 226	12, 972	13, 270	12, 522	177, 941

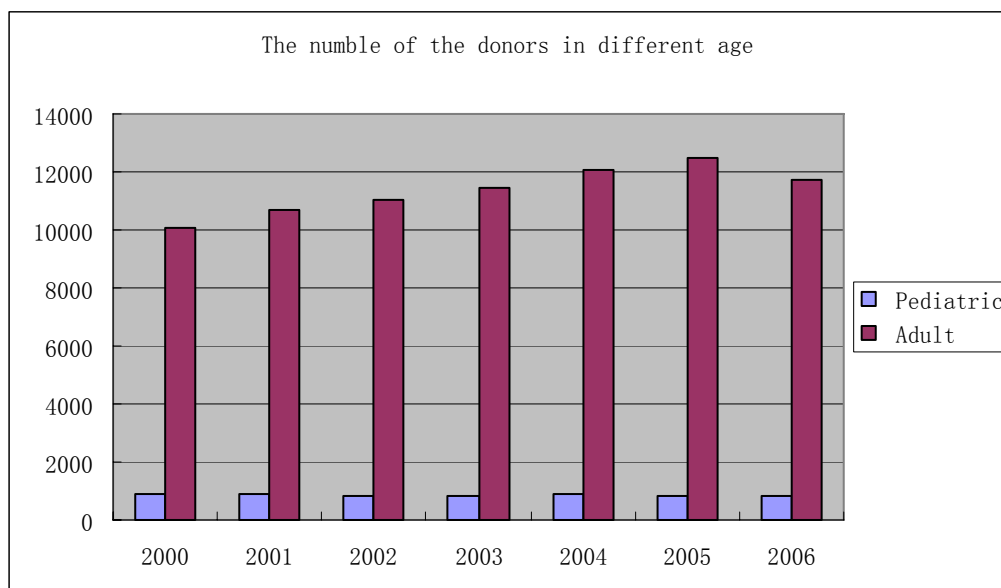


Figure 12: The comparison of the annual data in different age

From the figure 10, we can obtain that:

- The adult donations are much more than the kid donations. Generally speaking, few kids will donate their organs and the kid donations are mainly from the accident death of the kids.
- Meanwhile, the demand for the kid organ is not particularly large. The United Network for Organ Sharing Website recently (Oct 27, 2006) showed the age of waiting patients as: the donor under 18 counts just 748, while the number of the other donors older than 18 has reached 67, 544.

Although there is no controversy about the fact that kidney transplantation offers improved survival and quality of life in younger patients with end-stage renal failure, an ongoing debate exists about kidney transplantation in the elderly. Reduced mortality in transplanted patients compared to patients on the waiting list aged over 65 year, a prolonged waiting time in this patient subgroup significantly decreases the beneficial clinical outcome and the socio-economic advantages of early transplantation. Thus, every effort should be taken to reduce waiting time in this subgroup. An increased risk for post-transplant infections in the elderly has been shown and has to be considered in the selection of eligible patients. Besides, children under 18 years of age receive priority on the waiting list.

5) Ethnic group

Table 10: The annual number of the donors in different ethnicities

	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	To Date
White	6,355	6,483	6,649	7,215	7,386	8,036	8,337	8,530	8,566	9,048	9,170	8,697	130,815
Black	988	1,065	1,098	1,139	1,229	1,298	1,447	1,490	1,604	1,726	1,797	1,668	21,630
Hispanic	790	845	928	1,033	1,103	1,211	1,329	1,407	1,535	1,677	1,770	1,707	19,244
Unknown	19	68	28	43	22	2	0	1	16	0	0	0	317
Asian	131	146	183	201	217	292	297	280	309	345	354	304	3,545
AmericanIndian /Alaska Native	53	35	40	28	57	46	60	57	60	56	71	62	823
Pacific Islander	32	34	32	56	51	51	49	46	52	28	18	23	787
Multiracial	25	38	58	45	45	46	47	67	84	92	90	61	780
All Ethnicities	8,393	8,714	9,016	9,760	10,110	10,982	11,566	11,878	12,226	12,972	13,270	12,522	177,941

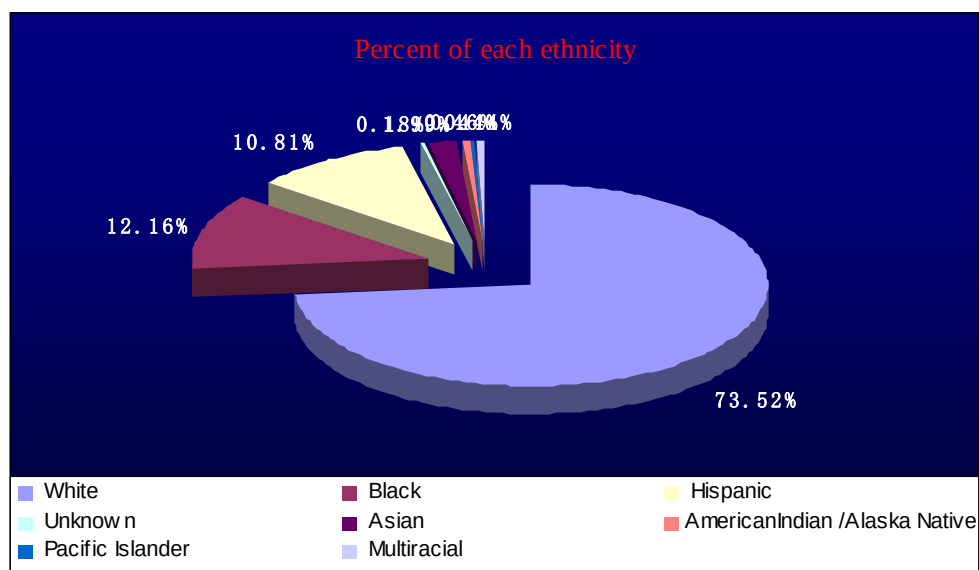


Figure 13: The percent of each ethnicity

A truth can be obtained from figure 11:

- The largest proportion of the donation is from the white (about 73.5%). Moreover, the growth of the total number is accelerating. The proportion from the Black and the Hispanic accounts about 22.9%. The donation from other races is few, and basically with no growth.
- The equality between different races should also be emphasized. Graft survival was analyzed using the Cox proportional hazards model. When analyzed by era, there was a temporal trend for a progressive decrease in the racial disparity in graft survival.

6) Geographic:

Table 11: The annual number of the donors in 11 regions

	2000	2001	2002	2003	2004	2005	2006	To Date
Region 1	213	242	226	218	233	242	224	3, 988
Region 2	707	707	749	730	822	859	817	12, 136
Region 3	863	944	875	883	992	1, 053	1, 003	14, 945
Region 4	474	493	499	538	596	577	562	8, 536
Region 5	719	736	794	837	899	993	1, 015	14, 127
Region 6	258	239	218	257	262	235	231	3, 827
Region 7	554	489	557	579	593	614	560	9, 375
Region 8	367	348	352	337	391	464	437	6, 585
Region 9	287	271	260	270	303	343	354	5, 347
Region 10	503	516	543	544	598	604	609	9, 434
Region 11	544	543	565	560	636	716	797	9, 520
All Regions	5, 489	5, 528	5, 638	5, 753	6, 325	6, 700	6, 609	97, 820

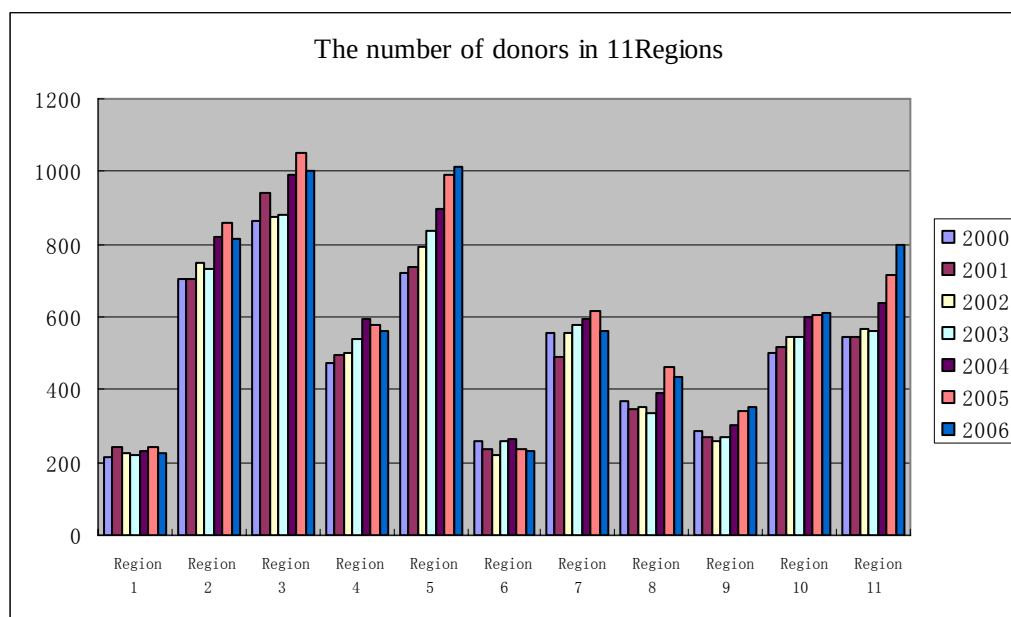


Figure 14: The comparison of the annual data in 11 regions

From the figure 12, we can find that:

- A slow increase of the donor number is clearly in every region we discussed, that is the number has remained stable in each region in some degree.
- Meanwhile there is a large difference between different districts. Therefore, Local patient is preferential. This may improve the use of the kidney recourse. Strategy can be drawn up according to the characteristics of each region.

Modifying the Models in Task 3 and Task 4

As our models in Task 3 and task 4 have mainly considered the kidney exchange problem in a non-biding status, but without taking the relevant ethic and policy issues into consideration. These are profound influence to the model 3 and the model 4 because of the constraints of the ethic and the policy. For example, considering when maximizing efficiency and optimize the quality of the, the religion or the race may brings an impact. That is to say, there may be a stop of the organ exchange due to the ethnic discrimination or the religion constraint. Another circs is that some patients make an unreasonable abandon when he is do a decision. It will cause a waste of organ resources, also is not allowed by the ethic and policy.

According to the above analysis, it is necessary to do some modification and adjustments of the models which built in task 3 and task 4. As these political and ethical issues are not related to the basic theories of the models, only add some constraints to the process of the match optimization and the decision-making of the patients. In view of various factors, we can add some relevant constraints to the each model.

For task 3

Firstly, there are some factors which will affect the medical and psychological dynamics of the situation of the patients that on the waiting list. We show them as follow: legislation, the partition of the transplant network, the government's economic input and the organ transplant technology.

- Legislation. Different law will make difference on the criteria which involved for getting on or coming off the list.
- The partition of the transplant network. The general selection way is the local patient first. So the partition of the transplant network will make impact on the patients that in different regions.
- The government's economic input. If the government plunges into more, there may be some difference on the policy. And it will directly affect the patients, improving the state of the kidney transplant.
- The organ transplant technology. Thinking that if the transplant technology is improved, the success rate of transplant will increase, and the waiting time may also shorten.

For task 4

There are also some main factors which will affect the decisions of the patients, including one's economic situation, faith, and education degree.

- Physical situation. The great physically shape is very important for a patient, a better physical situation will decrease the risk of the operation, and will improve the recovery after the operation.
- Economic situation. One's economic situation will serious affect one's decision on whether to take an offered kidney, for the cost of transplant is not low. And one also has pay for the recovery.
- Faith. Faith also has an impact on the decision-making, to different patients, they

may have different thoughts to the donors, and it will directly affect the patients' choice.

- Education degree. Distinct education degree also makes patients own different ideas about the risks that may appear in the surgery, alternative, and probabilities to get a better kidney if they reject the offered kidney.

Organ can be sold or not?

Background ^[14]

The therapeutic promise of transplanting organs from cadaver donors, as envisioned by the pioneers of transplantation, has never been realized because the demand for cadaver organs has far exceeded the supply. The waiting list for organ transplants continues to grow, and in 2006, nearly 8000 patients were removed from the list because of death. Consequently, many patients with end-stage organ failure are no longer relying solely on the waiting list. Instead, they are turning to spouses, friends, or strangers as possible donors—a medically acceptable alternative because advances in immunosuppression have eliminated the requirement of a genetic match for successful organ transplantation. In many U.S. transplantation centers, the number of kidneys transplanted from living donors has surpassed the number obtained from cadavers.

According to this phenomenon, there is a furious debate between the two kinds of people, that is whether the organ can be sold or not?

Advantage

Of course, some live donors may not feel very altruistic, but may be under strong family pressure to donate their organs to save the life of sibling or parent. And some non-relatives who are sufficiently altruistic to donate their organs are prevented from donating because of suspicions that the donors are being paid “under the table”.

Demand by wealthy individuals who finance much of their own transplant costs clearly also tends to be negatively related to costs. However, under the present system, many of these persons use the black market to pay either for live donors or for organs from cadavers in order to reduce their waiting times.

When donors or their heirs receive no compensation for their organs, one would not expect the supply of organs to depend much on the cost of the surgery. This is why the supply curve for either live or cadaver legal transplants is vertical, or independent of price.

If the organ can be traded, the organs which are get from the living person to be much more than before. Once a market for organs is in place, the actual demand would surpass present demand since now kidney transplants are only offered to people who have irreversible kidney failure. Other medical or surgical treatments for kidney problems are usually tried before consideration of a kidney transplant. So it is highly unlikely that the full demand for kidney transplants could be met with organs of deceased persons.

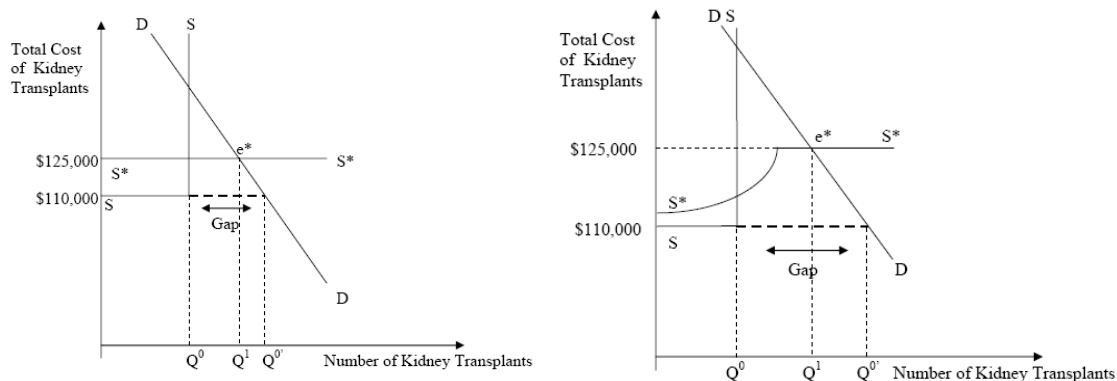


Figure 15, Figure 16: The cost of Kidney Transplant

This is shown in Figure 12, which plots the supply curve of kidney organs in a market with kidneys from both cadavers and live donors. This figure assumes a market with a single price for kidneys, no matter what the source, although our analysis suggests that organs from live donors are more valuable to recipients. For simplicity, the demand curve is taken to be the same as that in Figure 12.

The supply curve in this market would start just slightly above the cost of surgery since some cadaver organs would be made available cheaply. A rise in price would induce more cadaver organs to be offered, and perhaps even a few from live donors. Eventually, the available organs from cadavers would run out, and the supply price would rise sharply to reach the main market for live donors. At that point the supply elasticity rises sharply because the potential live donor market is huge relative to demand.

If our earlier discussion is right that the cadaver market would be insufficient to meet the demand for kidney transplants, the supply and demand curves in Figure 13 would intersect where supply is elastic, and where the marginal suppliers are live donors. Donors of organs from cadavers would receive a substantial surplus over the price they would be willing to accept because the market is too big to be supplied entirely by cadavers. The equilibrium price for organs would be similar to the price in Figure 12 which considers only live donors because live donors would be the marginal donors in full market equilibrium. Although most organs would be from cadavers, the market clearing price would be determined by the supply of live donors. Consequently, under the assumptions of this section about the number of cadaver organs available, the cost of buying organs to satisfy the demand for kidney transplants is not appreciably affected by integrating into the analysis the significant potential organ supply from cadavers.

The long wait for organs under the present system has encouraged the development of a black market in live or cadaver organs, where donors or their heirs get paid. These transplants are available only to wealthier individuals who usually must bear the total expense themselves. They are also often much riskier because organs are not screened as carefully for disease, are not matched as closely to recipients, and operating conditions and the quality of surgeons tend to be inferior. Allowing the purchase of organs would essentially knock out the black market in organs, and all its problems of quality control. The purchase of organs would also reduce the advantages of wealth in getting organs since poorer individuals in need of organs who cannot afford the black market would no longer have to wait so long before getting their

organs through the Medicaid or Medicare payment system.

Above all, the most effective answer to the viewpoint of paying for organs is that the present system imposes an intolerable burden on many very ill individuals who cannot afford to wait years until suitable organs become available. Increasing supply through payment would largely eliminate this wait and thus enormously improve the efficiency of the transplant market.

Disadvantage

However, if the organ business is permitted, there will be a series of behaviors which violate the ethics and moral.

One of the most common is that payment is “immoral” because it involves the “commodification” of body parts. Individuals who make this argument deny that people have the right to control the use of their bodies. Another reason why some critics have opposed payment for organs from cadavers is a fear that totalitarian governments and others might kill prisoners or other individuals to get their organs to sell. The sale of organs has also been opposed because poor individuals probably would be the main ones who would sell their organs. It is also claimed that payment for organ donations from living individuals would encourage impulsive and reckless provision of organs, partly because donors would not be able to sufficiently calculate the risks involved.

There are many other reason that we opposing the organ business, e.g. The people would sell one of their kidneys just to put food on the table; Some people might be for their organs; This may be course a new breed of “body-snatchers”; It would become an army of the poor.

One-page Report

——establishing an ethical market

Since then the number of organ transplants has grown rapidly, but not nearly as rapidly as the growth in the number of persons with defective organs that desire transplants. The result has been the development of longer and longer queues to receive organs. Seriously ill individuals often die while waiting for transplants. While people's lives continue to be put at risk by the dearth of organs available for transplantation, we must give urgent consideration to any useful propose that may make up the shortfall. Establishing a market for organ exchange is one such propose.

However, the revised Uniform Anatomical Gift Act of 1987, prohibited the sale of human organs. So the trading of the organ will act against the logistical constraint. Moreover, if the organ business is permitted, there will be a series of behaviors which violate the ethics and moral.

According to the above analysis, an ideal propose is to establish an ethical market for organ exchange. The market should be nonmonetary, ethically and supportable. And it must set up a regulated market system immediately.

The motives of living donors and the motives of families of deceased donors are complex and not necessarily purely altruistic. Families of deceased donors often regard organ donation as a way of giving meaning to the death or of allowing the person to "live on" in others. Because organ donation is voluntary and valuable, it should be considered a charitable gift. Society could explicitly thank organ donors for their gift, as is done with other charitable contributions, without jeopardizing its altruistic basis. New federal legislation should embrace ethically acceptable ways to encourage such charitable donation of organs, some of which are as follow: Donor Medal of Honor; Reimbursement for Funeral Expenses; Organ Exchanges; Medical Leave for Organ Donation; Donor Insurance; etc.

Brokering in the U.S according to market criteria of donor suitability would probably be no different. If the current prohibition against the sale of organs were rescinded, there would be little legal or ethical justification for preventing persons from bypassing the regulated system and using other means to obtain a better price for an organ from a more medically suitable donor. A federally regulated system would have to outlaw Internet bidding and set a controlled price for certain types of donors or continuously modify the price.

Here are some suggests for this market: Allowing noncash compensation, such as an income tax deduction or a reasonable payment for sales arranged through a national regulatory body; Prohibiting organ brokering by third parties; Imposing and monitoring strict donor selection criteria; Using the same organ allocation criteria used for cadaver organ distribution; Setting up a Incentives Payment Fund; etc.

In the final analysis, we believe that a market system of organ donation fosters class distinctions (and exploitation), infringes on the inalienable values of life and liberty, and is therefore ethically unacceptable. In contrast, non-monetary recognition of donation appeals to our notions of equity and, most important, does not subvert the altruistic social good that must be preserved in a revised system of organ donation. We urge Congress to retain the prohibition, established by the National Organ Transplant Act, against payment for organs in the U.S.

Task 6

Risks Analysis ^[19]

Nowadays, the risks — both physical and emotional — associated with kidney donation are more completely understood than they were in the early years of transplantation. Although donating may be an emotional necessity when the recipient is the donor's child, spouse, sibling, parent, or other loved one, this is far less likely to be the case for friends or anonymous donors. So when it comes to a decision-making, the potential donor may be affected by some risks from several aspects.

● **Probability success**

Before kidney donation, a number of physiological studies should be performed to establish that the donor is in excellent health and has normal renal function and that the anatomy of the kidney is suitable for transplantation, to assess the appropriateness of the donor's kidney for the recipient.

In addition to the team members who undertake the physical evaluation, transplantation teams generally include a psychiatrist who interviews potential donors to assess their motivation, who examines with them the psychosocial issues involved, and who supports them as they decide whether in fact to donate.

The decision to donate a kidney is inherently emotional. When all goes well, the donor almost always benefits. But the emotional stresses that will impact on some when donation fails occur, so the probability of success is a factor that affects one potential donor's decision.

The graft survival rates show in the following UNOS data for kidney transplants in the U.S (based on OPTN data as of 2006):

Table 12: Kidney Graft Survival Rates

Time after donation	From deceased donors	From living donors
One year	88.40%	94.40%
Three years	77.50%	87.60%
Five years	65.50%	78.70%

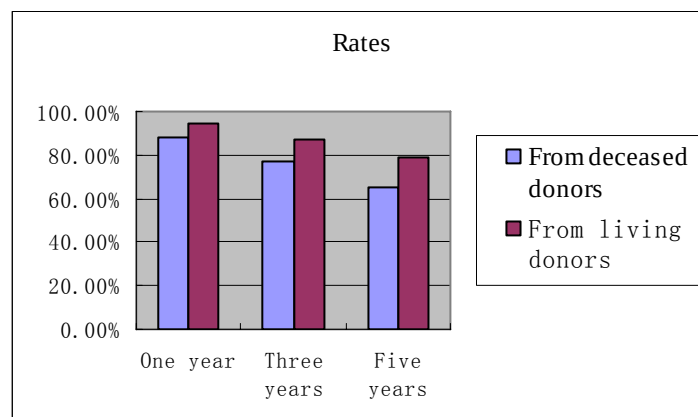


Figure 17: Kidney Graft Survival Rates

- **Perioperative risks**

Serious complications occur with major surgery such as nephrectomy, and perioperative mortality has been reported in about 1 out of 3000 living donor nephrectomies. The results have probably improved over time as for surgery in general. Minor complications are common. The rate of complications in this prospective series appears to be somewhat higher than before.

Recovery after surgery is faster with laparoscopic nephrectomy, and reports so far also clearly show that post-operative pain related to nephrectomy is reduced. However, the risk of serious acute complications and bleeding may be increased.

- **Long-term mortality**

Although increased long-term mortality has not been reported in healthy persons after nephrectomy, an increased survival may not be surprising as the donors are positively selected and screened for disease. Thus, there are no indications that kidney donation has any negative impact on long-term survival.

- **Long-term morbidity**

Hypertension and kidney impairment are potential risks associated with kidney donation. According to the earlier studies found little or no effect on blood pressure. However, there was an average increase in blood pressure of 2–3 mmHg and a further increase in systolic pressure of 1 mmHg for each decade following kidney ablation^[20].

Glomerular filtration rate (GFR) is obviously reduced following nephrectomy. Although most centres find the risk of kidney donation acceptable, the risk should not be neglected. Donors with low GFRs may be at particular risk and graft survival of the recipients is decreased with GFRs under 80 ml/min. Relatives of recipients with end-stage kidney disease may also be genetically susceptible for kidney disease, for example diabetes or SLE or for progression of such disease.

- **Quality of life and disability**

Systematic reports on the effects of kidney donation on quality of life and psychosocial function long-term were first published in the 1990s. Overall, the quality of life was significantly better than for the background population. Similar positive results have recently been reported from Japan, albeit in a smaller series. All centres concluded in their reports that the results were overwhelmingly positive and that the donors had a better quality of life than the background population. This is not surprising considering the selection process of donors with excellent physical and mental health status.

Model design—Two-Box decision Mode

By the fuzz mathematics theory , we bring in two boxes to deal with the uncertain factors discussed above, the first box is built for the objective risk factors, including the probability of success for the recipient, the probability of survival of donors , the probability of future health problems and the postoperative pain and recovery. The other box is used to manage the

subjective factors (personal issues), with the phobias, irrational fears, misinformation, previous experience, level of the altruism and trust. Thus for a potential donor, we can establish a Two-boxes to help him make the decision whether to offer his kidney.

Obtain the relevant data from the internet, as for the factors that have no data resource, we can quantify the impact of these factors. By the standardization of data processing, we can get the objective input part of the Two-Box model. The object factors about the personal issues can be dealt with in the same way.

Once the value of each factor is defined, we consider how our Two-Box model will work as follows:

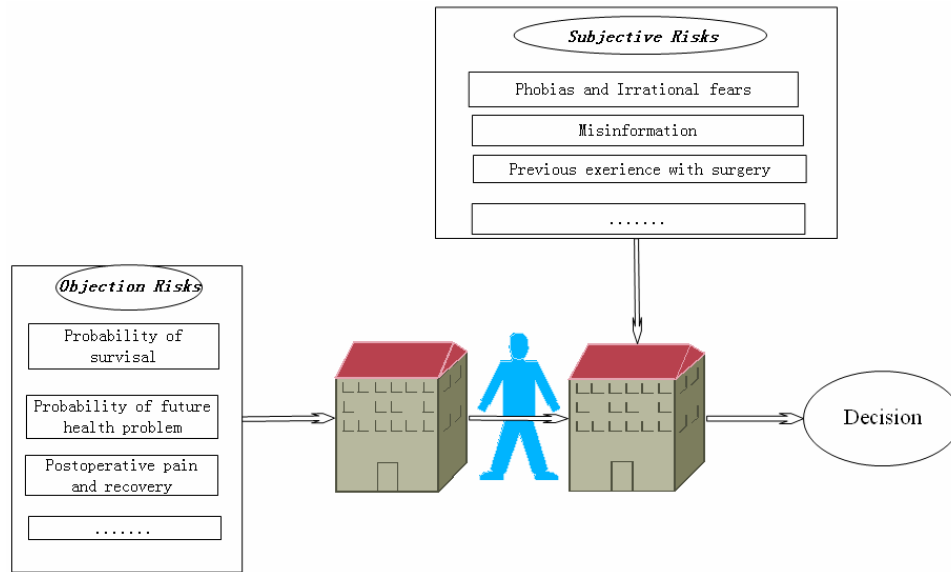


Figure 18: the Two-Box decision model sketch map

Definition and notation

Let we note a_i be the i_{th} objective factor, and b_j the j_{th} subjective factor. For each a_i or b_j , we suppose that the potential donor have to make the decision only when $a_i \in [a_0, a']$, $b_j \in [b_0, b']$. While $a_i < a_0$, we suppose that the risk factor is certainly acceptable. While $a_i > a'$ we suppose that he risk factor is certainly unacceptable and the person will certainly reject to donate his kidney. The situation is the same with the personal issue factors. Then for each a_i or b_j , we define a factor weight ω_{a_i} or ω_{b_j} .

The decision function

We assume that the risk value is a logarithmic function of $\frac{a' - a_0}{a_i - a_0}$ and the personal factors' influence value is a linear function of $\frac{b' - b_0}{b_i - b_0}$, then we can obtain the combined

Decision function:
$$F(p) = \omega_1 \sum_i \omega_{ai} \times \ln \frac{a' - a_0}{a_i - a_0} + \omega_2 \sum_j \omega_{bj} \times \frac{b' - b_0}{b_i - b_0}$$

P stands for the potential donor person. ω_1 , ω_2 are the weight of the two boxes.

For a standard parameter F^* , $F(p) < F^*$, P donates the kidney, otherwise rejects the donation.

The weight ω_{ai} or ω_{bj} and F^* can be modified for a better decision result.

Modify Models

First we review the model in task 3, in that model we only consider the medical and psychological situation from patients' perspective, but when it comes to the more practical case, we have to take into account the donors' decision-making.

In model 3, there is a set of compatible donors for each patient, and we construct the preference ordering base on the utility function. But in the utility function, we don't consider the donor's situation. But when we want to maximize the number and quality of exchanges, the donor's health should be treated seriously. Taking the donor's risks, including the probability of survival, the probability of future health, the postoperative pain and recovery and even the perceived risks and personal issues into account, we should modify the utility function to make it more reasonable. At the same time, the donor's probability to donate also has an impact on the utility function. As the utility function is modified, the preference list to every patient will be reordered too.

For the model in task 4, we devise a strategy for a patient to decide whether to take an offered kidney or not, in that model we mainly consider the quality of the kidney from the donor and the health state of the patient, from the patients' perspective. But here we have to think over the probability of the donor to donate. To one offered kidney, besides the quality and health state, the patient also should consider the risks which may affect the donor's decision from the donor's perspective. So the equations in the model should be changed to the ones that containing the probability of the donor's donate.

The network size effect

When the potential donor enters a list paired network rather than a direct transplant to the relative or friend, the size n of the n -paired network may has some effect on our models.

As it is discussed in task 3, a larger size n may guarantee a higher quality and number of the kidney exchange. But when it comes to the decision of the potential donor, we think the increase of the size n may result in a negative effect. This is because when the n -paired network becomes larger, the exchange becomes complex and then the objective risk for the donor may increase in some degree. The following changes with phobias, irrational fears, level of altruism and level of trust would turn to a bad state.

Appeal to more donors

Organs for transplantation are usually obtained from living or cadaver donors. Unfortunately, these sources have so far been unable to keep up with demand. As a result, there are a large and steadily increasing number of potential awaiting transplantation, some of whom will die before an organ can be found. In an attempt to improve this tragic situation, several solutions have been proposed. This review will consider proposals designed to increase the availability of human organs without involving to commercialism as follow.

- Expand the use of living donors by: 1) Encouraging donations by genetic relatives; 2) Allowing volunteers a greater voice in determining their own suitability; 3) Encouraging the use of emotionally related individuals and accepting altruistic strangers; 4) Considering motivated identical twin minors and older adolescents as donors; etc.
- Suggestions for increasing the pool of cadaver donors include: 1) Overcoming the family consent barrier by presuming consent, mandating completion of binding advanced directives, or by eliminating the need for consent entirely; 2) Reconsidering non-heart-beating donors; 3) Elective ventilation for organ donation; and 4) Accepting organs from anencephalic infants before brain death occurs; etc.
- Written information for potential living kidney donors

To meet the constantly growing demand for organs for transplantation the use of living related and unrelated donors continues to increase. Transplant units with a living-donor program often provide written information to their potential kidney donors. We saw a need to assess the contents of these brochures. Written information for potential live kidney donors was requested from different Transplant units throughout the world.

Here we points at several essential requirements for the contents, which should be fulfilled to enable the potential donor to make a thorough decision. We suggest the following guidelines for written information:

- Voluntarism-the fact that donation should be an act of free will
- Medical suitability and Risk of graft loss
- Benefits for the recipient with a live kidney donor
- The assessment, and the estimated time this will take
- Postoperative course and the expected time
- Short-term donor risks—medical and psychological
- Long-term donor risks—medical and psychological
- Sick-leave duration
- Financial and Legal conditions

The information for potential donors should be given separately from general information of transplantation. The language should be easy to understand for most people and rather informal. Use illustrations and wide line spacing to catch the reader's interest.

All of these proposals raise concerns which are discussed. Those approaches considered to be ethically acceptable and to hold promise for success should be vigorously pursued, beginning with carefully designed pilot studies. Hopefully, such an approach will eventually increase the number of organs available for patients suffering from end-stage organ disease.

Strengths and Weaknesses

Strength

- The Monte Carlo simulation has been perfectly used in our models, and the simulation results are consistent with the reality.
- In task 3, we introduced a preference construction mechanism (R-C) in order to improve the exchange quality. The R-C mechanism has taken both consideration of the medical and psychological dynamic of the situation. The chain rules can also modified in a degree.
- The models used in our paper is promotional, in view of different consideration, we can modify our models conveniently.

Weakness

- Some special data can't be found, and it makes that we have to do some proper assumption before the solution of our models. A more abundant data resource can guarantee a better result in our models.

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Appendix

Appendix I

```
#include <stdio.h>
#include <math.h>
#include <memory.h>
#include <stdlib.h>
#include <time.h>
#include <algorithm>
using namespace std;

/* define the bloodtype */
#define bloodtypeO 0
#define bloodtypeA 1
#define bloodtypeB 2
#define bloodtypeAB 3

#define NDONOR 10000
#define NPATIENT 40000

/* define the Variables */
int Sum;
int SumDonor;
int SumPatient;
int Match;
int RandomState[11];
int RandomP[11];

struct NodeDonor {
    int bloodtype, region;
    int HLA;
    bool Used;
};

struct NodePatient {
    int bloodtype, region;
    int HLA;
    int Point;
};

struct Region {
    int nDonor;
    NodeDonor Donor[2000];
```

```
    int npatient;
    NodePatient patient[8002];
}State[11];

double DonoteRate[11] = {
    0.040768759,
    0.124064608,
    0.152780617,
    0.087262319,
    0.144418319,
    0.039122879,
    0.095839297,
    0.067317522,
    0.054661623,
    0.096442445,
    0.097321611
};

bool BloodMatch[4][4] = {
    1, 0, 0, 0,
    0, 1, 0, 1,
    0, 0, 1, 0,
    0, 0, 0, 1
};

double BloodRate[4] = {
    0.487783985,
    0.330837989,
    0.144040968,
    0.037337058
};

bool cmp(NodePatient p1, NodePatient p2)
{
    return p1.Point > p2.Point;
}

int Rondon(int min, int max)
{
    int n;
    return rand() % (max - min + 1) + min;
}

void Initial()
```

```

{
    srand(time(NULL));

    int i;
    for(i = 0; i < 11; i++)
    {
        State[i].nDonor = 0;
        State[i].npatient = 0;
    }
    SumDonor = 0;
    SumPatient = 0;
    for(i = 0; i < 11; i++)
    {
        RandomState[i] = (int)(5000.0 * DonoteRate[i] * (rand() % 2 + 1));
        RandomP[i] = RandomState[i] * (rand() % 2 + 3);
        SumDonor += RandomP[i];
        SumPatient += RandomState[i];
    }
    for(i = 1; i < 4; i++)
    {
        BloodRate[i] += BloodRate[i-1];
    }
    Sum = 0;
    Match = 0;
}

void MonteCarloSimulate()                /* Monte Carlo Simulate */
{
    int CountDonor = 0;
    int CountPatient = 0;
    bool flag;
    NodeDonor TempDonor;
    NodePatient TempPatient;
    int i, j, n;

    while(1)                            /* randomly create donors and patients */
    {
        n = rand() % 11;
        if(State[n].nDonor >= RandomState[n] && State[n].npatient >= RandomP[n])
        {
            j = (n + 1) % 11;
            while(j != n)
                if(State[j].nDonor >= RandomState[j] && State[j].npatient >= RandomP[j]) j++; j %=
11;

```

```

        else break;
    if(j == n) break;
    else n = j;
}

if(State[n].npatient < RandomP[n] && rand()%2)
{
    TempPatient.bloodtype = rand() % 4;
    TempPatient.region = n;
    TempPatient.HLA = rand() % 128;
    TempPatient.Point = 1;

    int Best = 0;
    int min = 1000;
    for(i = 1; i <= State[n].nDonor; i++)
    {
        if(State[n].Donor[i].Used
BloodMatch[ State[n].Donor[i].bloodtype ][ TempPatient.bloodtype ])
        {
            if(abs(State[n].Donor[i].HLA - TempPatient.HLA) <= 8)
            {
                if(abs(State[n].Donor[i].HLA - TempPatient.HLA) < min)
                {
                    Best = i;
                    min = abs(State[n].Donor[i].HLA - TempPatient.HLA);
                }
            }
        }
    }
    if(Best)
    {
        State[n].Donor[Best].Used = false;
        Sum++;
        Match++;
        RandomP[n]--;
    }
    else
    {
        for(j = 0; j < 11; j++)
        {
            if(j == n) continue;
            min = 1000;
            for(i = 1; i <= State[j].nDonor; i++)
            {

```

```

        if(State[j].Donor[i].Used
BloodMatch[ State[j].Donor[i].bloodtype ][ TempPatient.bloodtype ])
        {
            if(abs(State[j].Donor[i].HLA - TempPatient.HLA) <= 8)
            {
                if(abs(State[j].Donor[i].HLA - TempPatient.HLA) < min)
                {
                    Best = i;
                    min = abs(State[j].Donor[i].HLA - TempPatient.HLA);
                }
            }
        }
    }
    if(Best && min < 1000)
    {
        State[j].Donor[Best].Used = false;
        Sum += 1;
        Match++;
        RandomP[n]--;
        break;
    }
}
if(Best == 0)
{
    State[n].npatient++;
    State[n].patient[ State[n].npatient ] = TempPatient;
    for(i = 1;i < State[n].npatient; i++)
    {
        if(State[n].patient[i].Point > 0)
            State[n].patient[i].Point++;
    }
}
}
else if(State[n].nDonor < RandomState[n])
{
    TempDonor.bloodtype = rand() % 4;
    TempDonor.region = n;
    TempDonor.HLA = rand() % 128;
    TempDonor.Used = true;

    sort(State[n].patient + 1, State[n].patient + State[n].npatient + 1, cmp);

    int Best = 0;

```



```

        for(i = 1; i <= State[n].npatient; i++)
        {
            if(State[n].patient[i].Point > 0 &&
BloodMatch[ TempDonor.bloodtype ][ State[n].patient[i].bloodtype ])
            {
                if(abs(State[n].patient[i].HLA - TempDonor.HLA) <= 8)
                {
                    Best = i;
                    break;
                }
            }
            else if(State[n].patient[i].Point == 0) break;
        }
        if(Best)
        {
            Sum ++;
            State[n].patient[Best].Point = 0;
            Match++;
            RandomState[n]--;
        }
        else
        {
            State[n].nDonor++;
            State[n].Donor[ State[n].nDonor ] = TempDonor;
        }
    }
}

void Output()
{
    printf("Match    : %d\n", Match);
    printf("SumCost : %d\n", Sum);
    printf("%d    %d\n", SumDonor, SumPatient);
}

int main()
{
    Initial();
    MonteCarloSimulate();
    Output();
    return 0;
}

```

Appendix II

```
#include <iostream>
#include <stdlib.h>
#include <time.h>
#include <memory.h>
#include <algorithm>
#include <math.h>
using namespace std;

#define MaxPatient 300
#define MaxDonor    300
#define MaxPairs    300

#define BloodTypeO   0
#define BloodTYpeA   1
#define BloodTypeB   2
#define BloodTypeAB  3

#define Female 0
#define Male   1

#define AgeLow  18
#define AgeHigh 79

#define rational 0
#define cautious 1

#define WaitListOption 0.3
#define w1 0.5
#define w2 0.5

struct node {
    int bloodtype;
    int HLA[6];
    int gender;
    int age;
} patient[MaxPatient + 2], donor[MaxDonor + 2];

struct Preference {
    int number;
    double prefer_point;
} pre[MaxPatient + 2][MaxDonor + 2];
```

```
bool cmp(Preference donor1, Preference donor2)
{
    return (donor1.prefer_point > donor2.prefer_point) ||
        (donor1.prefer_point == donor2.prefer_point && donor1.number < donor2.number);
}
```

```
struct edge {
    int x, y;
    double pri;
};
```

```
bool operator < (edge e1, edge e2) {
    if(e1.pri < e2.pri)    return true;
    return false;
}
```

```
bool Related[MaxPairs];
double AvgTrans;
double AvgOwnTrans;
double AvgHLAmis;
double AvgRisk;
```

```
int AgePartion[6] = {
    0, 18, 34, 49, 64, 79
};
```

```
bool compatible[4][4] = {
    1, 1, 1, 1,
    0, 1, 0, 1,
    0, 0, 1, 1,
    0, 0, 0, 1
};
```

```
double Patient_BloodType_Frequency[4] = {
    0.456,  // O
    0.395,  // A
    0.111,  // B
    0.038   // AB
};
```

```
double Patient_Gender_Frequency[2] = {
    0.409,  // female
    0.591   // male
};
```

```
double Patient_Age_Frequency[5] = {
    0.056, // < 18
    0.13,  // 18 - 34
    0.349, // 35 - 49
    0.389, // 50 - 64
    0.076 // > 64
};

double Unrelated_Living_Donors_Frequency[2] = {
    0.535, // spouse
    0.465  // other
};

double Donor_Age_Frequency[5] = {
    0.056, // < 18
    0.13,  // 18 - 34
    0.349, // 35 - 49
    0.389, // 50 - 64
    0.076 // > 64
};

double Positive_Crossmatch__Frequency[2] = {
    0.333, // female patient - husband
    0.111  // other
};

void Init()
{
    srand(time(NULL));
    int i;
    for(i = 1; i < 4; i++) Patient_BloodType_Frequency[i] += Patient_BloodType_Frequency[i-1];
    for(i = 1; i < 5; i++) Patient_Age_Frequency[i] += Patient_Age_Frequency[i-1];
    for(i = 1; i < 5; i++) Donor_Age_Frequency[i] += Donor_Age_Frequency[i-1];
}

void SubInitial()
{
    memset(Related, false, sizeof(Related));
    AvgTrans = 0;
    AvgHLAmis = 0;
    AvgRisk = 0;
    AvgOwnTrans = 0;
}
```

```
int HLA_mismatch(int HLA_a[], int HLA_b[])
{
    int i, total = 0;
    for(i = 0; i < 6; i++)
    {
        total += abs(HLA_a[i] - HLA_b[i]);
    }
    return total;
}
```

```
double Cal_Preference(node patientI, node donorI)
{
    double preference = 0;
    double a, b;
    if(patientI.age < 60) a = 1.06, b = 1.12;
    else a = 1.05, b = 1.10;
    preference = pow(a, 1.0 * HLA_mismatch(patientI.HLA, donorI.HLA)) * pow(b, donorI.age / 10);
    return preference;
}
```

```
double Estimate_Preference(node patientI, node donorI)
{
    double preference = 0;
    double a, b;
    double avg_HLA = 3.2;
    double avg_age = 50;
    a = w1 * (1.0 * HLA_mismatch(patientI.HLA, donorI.HLA) - avg_HLA) / avg_HLA;
    b = w2 * (1.0 * donorI.age - avg_age) / avg_age;
    return a + b;
}
```

```
void Random()
{
    int i, j;
    double RandRealNumber;
    for(i = 1; i <= MaxPairs; i++)
    {
        RandRealNumber = 1.0 * (rand() % 10000) / 10000;
        for(j = 0; j < 4; j++) if(RandRealNumber < Patient_BloodType_Frequency[j]) break;
        patient[i].bloodtype = j;

        RandRealNumber = 1.0 * (rand() % 10000) / 10000;
        patient[i].gender = RandRealNumber > 0.409 ? 1 : 0;
    }
}
```

```

    patient[i].age = rand() % (AgeHigh - AgeLow + 1) + AgeLow;
    for(j = 0; j < 6; j++)
    {
        patient[i].HLA[j] = rand() % 2;
    }

    donor[i].bloodtype = rand() % 4;
    donor[i].gender = rand() % 2;
    donor[i].age = rand() % (AgeHigh - AgeLow + 1) + AgeLow;
    for(j = 0; j < 6; j++)
    {
        donor[i].HLA[j] = rand() % 2;
    }

    RandRealNumber = 1.0 * (rand() % 10000) / 10000;
    if(RandRealNumber < Unrelated_Living_Donors_Frequency[0])
    {
        Related[i] = true;
    }
}

void NoExchange()
{
    int i;
    double Risk = 0;
    double HLAamis = 0;
    double TotalTrans = 0;
    for(i = 1; i <= MaxPairs; i++)
    {
        if(compatible[ donor[i].bloodtype ][ patient[i].bloodtype ])
        {
            HLAamis += HLA mismatch(patient[i].HLA, donor[i].HLA);
            TotalTrans += 1;
            Risk += Cal_Preference(patient[i], donor[i]);
        }
    }
    AvgTrans += TotalTrans / MaxPairs;
    AvgHLAamis += HLAamis / TotalTrans;
    AvgRisk += Risk / TotalTrans;
}

int Pair[MaxPairs][2];

```

```
void Paired()
{
    int i, j;
    int countpair = 0;
    bool Pused[MaxPatient + 2];
    bool Dused[MaxDonor + 2];
    memset(Pair, false, sizeof(Pair));
    memset(Pused, false, sizeof(Pused));
    memset(Dused, false, sizeof(Dused));
    for(i = 1; i <= MaxPairs; i++)
    {
        if(compatible[ donor[i].bloodtype ][ patient[i].bloodtype ])
        {
            Pused[i] = true;
            Dused[i] = true;
            countpair++;
            Pair[countpair][0] = i;
            Pair[countpair][1] = i;
        }
    }

    double Risk = 0;
    double HLAamis = 0;
    double TotalTrans = countpair;

    bool change = true;
    while(change)
    {
        change = false;
        for(i = 1; i <= MaxDonor; i++)
        {
            for(j = 1; j <= MaxPatient; j++)
            {
                if(compatible[ donor[i].bloodtype ][ patient[j].bloodtype ] && !Pused[i] && !Dused[j])
                {
                    Pused[i] = true;
                    Dused[j] = true;
                    countpair++;
                    Pair[countpair][0] = i;
                    Pair[countpair][1] = j;
                    change = true;
                }
            }
        }
    }
}
```

```

    }
}

for(i = 1; i <= countpair; i++)
{
    HLAmis += HLAismatch(patient[Pair[i][0]].HLA, donor[Pair[i][1]].HLA);
    Risk += Cal_Preference(patient[Pair[i][0]], donor[Pair[i][1]]);
}

AvgOwnTrans += TotalTrans / MaxPairs;
TotalTrans = countpair;
AvgTrans += TotalTrans / MaxPairs;
AvgHLAmis += HLAmis / TotalTrans;
AvgRisk += Risk / TotalTrans;
}

int graph[MaxPatient + 2];
bool Pused[MaxPatient + 2];
int q[MaxPatient * 2];
int match[MaxPatient + 2];

void TTCC()
{
    memset(match, 0, sizeof(match));
    memset(graph, 0, sizeof(graph));
    memset(Pused, false, sizeof(Pused));
    int i, j, k;
    for(i = 1; i <= MaxPatient; i++)    // preference construction
    {
        for(j = 1; j <= MaxDonor; j++)
        {
            pre[i][j].number = j;
            if(compatible[ donor[j].bloodtype ][ patient[i].bloodtype ])
                pre[i][j].prefer_point = Cal_Preference(patient[i], donor[j]);
            else
                pre[i][j].prefer_point = 0;
        }
        sort(pre[i] + 1, pre[i] + MaxDonor + 1, cmp);
    }

    while(1)    // create the direct graph
    {
        for(i = 1; i <= MaxPatient; i++)
        {

```



```
        if(!Pused[i]) break;
    }
    if(i > MaxPatient) break;

    graph[i] = pre[i][1].number;
    k = graph[i];
    Pused[i] = true;

    while(!Pused[k])
    {
        graph[k] = pre[k][1].number;
        Pused[k] = true;
        k = graph[k];
    }
}

bool used[MaxPatient];
memset(used, false, sizeof(used));
memset(Pused, false, sizeof(Pused));
bool change = true;

int countpair = 0;

while(change)                                // using TTCC to simulate the procedure
{
    change = false;
    bool loop = false;
    for(i = 1; i <= MaxPatient; i++) // find the circle
    {
        if(Pused[i]) continue;
        memset(used, false, sizeof(used));
        k = i;
        used[k] = true;
        while(!Pused[ graph[k] ] && !used[ graph[k] ])
        {
            used[graph[k]] = true;
            k = graph[k];
            if(k == i) {loop = true; break;}
        }
        if(loop)
        {
            change = true;
            while(graph[k] != i && !Pused[ graph[k] ])
            {
```

```
        match[ graph[k] ] = k;
        Pused[ graph[k] ] = true;
        k = graph[k];
    }
    Pused[i] = true;
    match[i] = k;
    break;
}
}

if(!change)
{
    int l = 0;
    int c = 0;
    int flag;
    for(i = 1; i <= MaxPatient; i++)
    {
        if(Pused[i]) continue;
        k = i;
        c = 0;
        memset(used, false, sizeof(used));
        used[i] = true;
        while(!Pused[ graph[k] ] && !used[ graph[k] ])
        {
            k = graph[k];
            used[k] = true;
            c++;
            if(k == i) break;
        }
        if(c > 1) {l = c; flag = i;}
    }
}
```

```
if(l) change = true;
k = flag;
used[k] = true;
Pused[k] = true;
while(!Pused[ graph[k] ] && !used[ graph[k] ])
{
    used[ graph[k] ] = true;
    match[ graph[k] ] = k;
    Pused[ graph[k] ] = true;
    k = graph[k];
}
```

```

    }
}

double estimate;

double Risk = 0;
double HLAmis = 0;
double TotalTrans = MaxPatient;

int wait = 0;
for(i = 1; i <= MaxPatient; i++)
{
    if(match[i])
    {
        estimate = Cal_Preference(patient[i], donor[match[i]]);
        if(estimate < 1.308977)
        {
            double r = 1.0 * (rand() % 10000) / 10000;
            if(r <= 0.3) continue;
            if(match[i] == i) countpair++;
            Risk += Cal_Preference(patient[i], donor[match[i]]);
            HLAmis += HLA mismatch(patient[i].HLA, donor[match[i]].HLA);
        }
        else
        {
            TotalTrans -= 1;
            wait++;
        }
    }
    else
    {
        Risk += 0.8 * Cal_Preference(patient[i], donor[i]);
        HLAmis += 0.8 * HLA mismatch(patient[i].HLA, donor[i].HLA);
        match[i] = i;
    }
}

double s = MaxPairs;

AvgTrans += TotalTrans / MaxPatient;
AvgOwnTrans += 1 - AvgTrans;
AvgHLAmis += HLAmis / MaxPatient;
AvgRisk += Risk / MaxPatient;
}

```

```
int main()
{
    Init();
    SubInitial();
    for(int i = 1; i <= 100; i++)
    {
        Random();
        NoExchange();
    }
    printf("%.2lf%%    %.2lf    %.2lf    %.2lf%%\n", AvgTrans , AvgHLAmis/100, AvgRisk/100,
AvgOwnTrans);

    SubInitial();
    for(int i = 1; i <= 100; i++)
    {
        Random();
        Paired();
    }
    printf("%.2lf%%    %.2lf    %.2lf    %.2lf%%\n", AvgTrans , AvgHLAmis/100, AvgRisk/100,
AvgOwnTrans);

    SubInitial();
    for(int i = 1; i <= 1; i++)
    {
        Random();
        TTCC();
    }
    printf("%.2lf%%    %.2lf    %.2lf    %.2lf%%\n", AvgTrans*100 , AvgHLAmis, AvgRisk,
AvgOwnTrans*100);
    return 0;
}
```